

EE115B *Digital Circuits*

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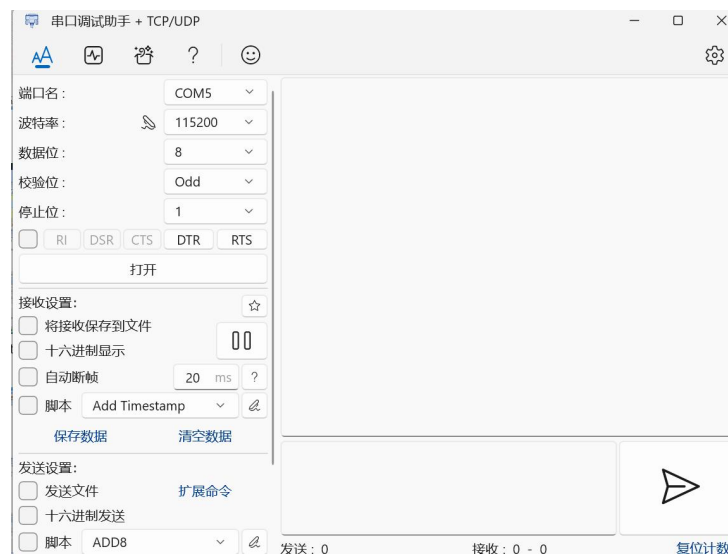


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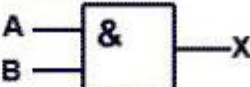
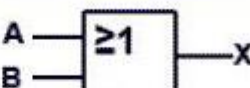
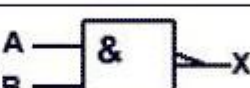
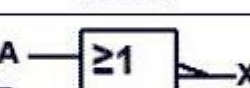
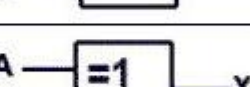
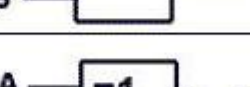
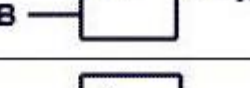
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- BCD 8421 code
- Gray code
- ASCII code
- Parity method for error detection

Decimal	Binary Code	Gray Code	Decimal	Binary Code	Gray Code
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000



The actions of any of these gates can also be described using Boolean statements.

ANSI Symbol	IEC Symbol	NAME
		AND
		OR
		NAND
		NOR
		XOR
		XNOR
		NOT

Boolean	NAME
$X = A \cdot B$	AND
$X = A + B$	OR
$X = \overline{A \cdot B}$	NAND
$X = \overline{A + B}$	NOR
$X = A \oplus B$	XOR
$X = \overline{A \oplus B}$	XNOR
$X = \overline{A}$	NOT

Fig. 2.1.1 ANSI & IEC Gate Symbols

Bit Manipulation

Set Bit: Or 1

Clear Bit: And 0

Invert Bit: Not, XOR 1

Operators	Description	Example	Result
	Bitwise OR	4 2	6
&	Bitwise AND	1&3	1
^	Bitwise exclusive OR	4^3	7
<<	Bitwise Left Shift	2<<1	4
>>	Bitwise Right Shift	4>>1	2
~	Bitwise Not	~6	-7

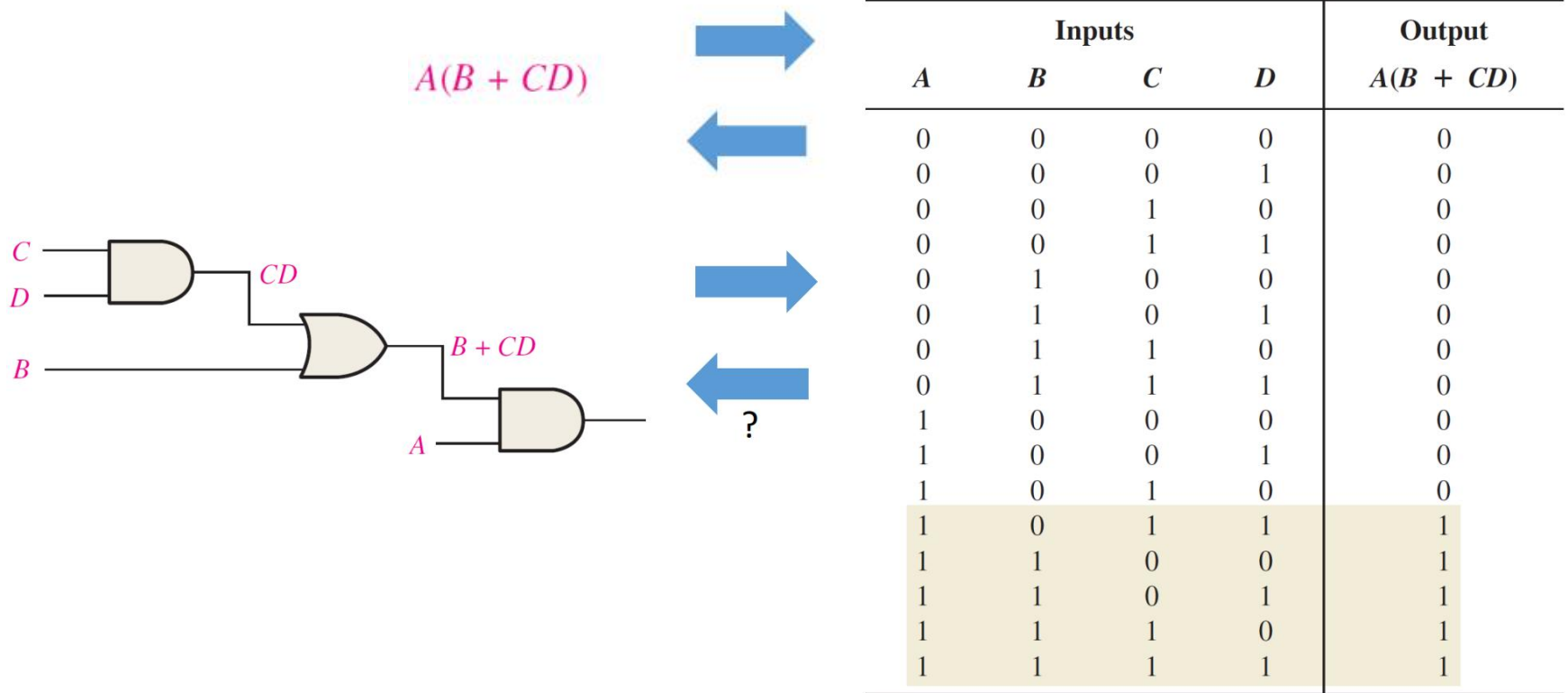


From Basic Primitives to
Practical Calculations:

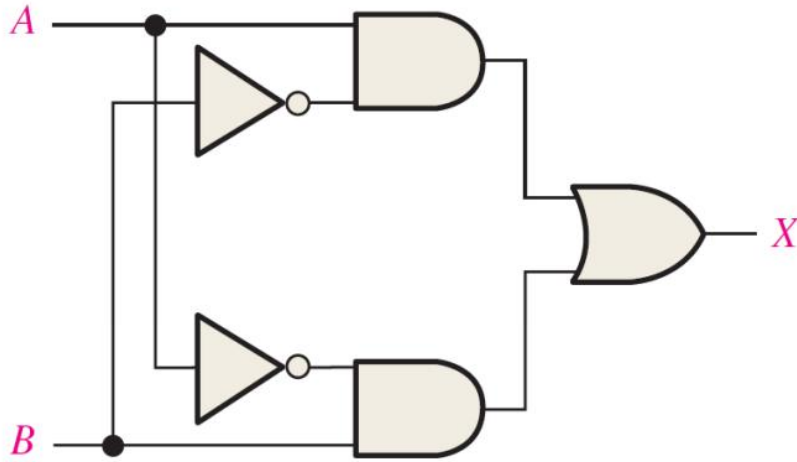
Boolean Expression &
Logic Circuit

Boolean Expression & Truth Table & Logic Circuit

A logic function can have different representations.

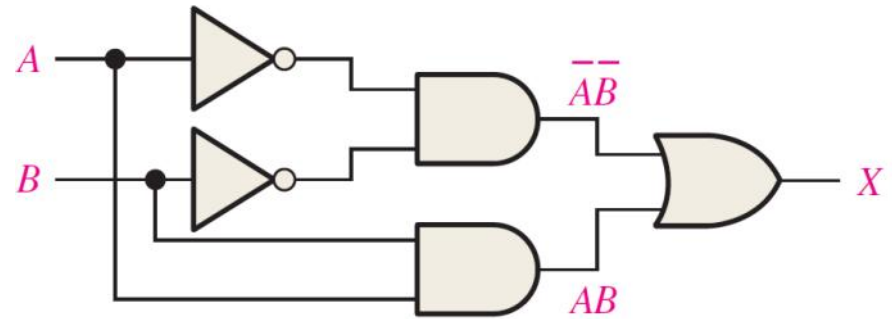


Exercise: Write down boolean expression



XOR

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



XNOR

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1



Commutative Laws

The **commutative laws** are applied to addition and multiplication. For addition, the commutative law states

In terms of the result, the order in which variables are ORed makes no difference.

$$A + B = B + A$$

For multiplication, the commutative law states

In terms of the result, the order in which variables are ANDed makes no difference.

$$AB = BA$$



Associative Laws

The **associative laws** are also applied to addition and multiplication. For addition, the associative law states

When ORing more than two variables, the result is the same regardless of the grouping of the variables.

$$A + (B + C) = (A + B) + C$$

For multiplication, the associative law states

When ANDing more than two variables, the result is the same regardless of the grouping of the variables.

$$A(BC) = (AB)C$$

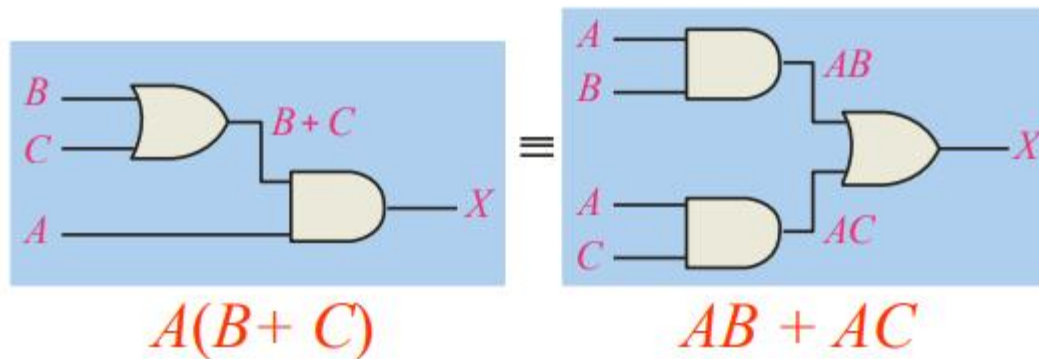


Distributive Law

The **distributive law** is the factoring law. A common variable can be factored from an expression just as in ordinary algebra. That is

$$AB + AC = A(B + C)$$

The distributive law can be illustrated with equivalent circuits:



Rules of Boolean Algebra

1. $A + 0 = A$

2. $A + 1 = 1$

3. $A \cdot 0 = 0$

4. $A \cdot 1 = A$

5. $A + A = A$

6. $A + \bar{A} = 1$

7. $A \cdot A = A$

8. $A \cdot \bar{A} = 0$

9. $\overline{\bar{A}} = A$

10. $A + AB = A$

11. $A + \bar{A}B = A + B$ *

12. $(A + B)(A + C) = A + BC$ *



Rules of Boolean Algebra

- Rule 1: $A + 0 = A$, A variable ORed with 0 is always equal to the variable



- Rule 2: $A + 1 = 1$, A variable ORed with 1 is always equal to 1.



- Rule 3: $A \cdot 0 = 0$, A variable ANDed with 0 is always equal to 0.



- Rule 4: $A \cdot 1 = A$, A variable ANDed with 1 is always equal to the variable.

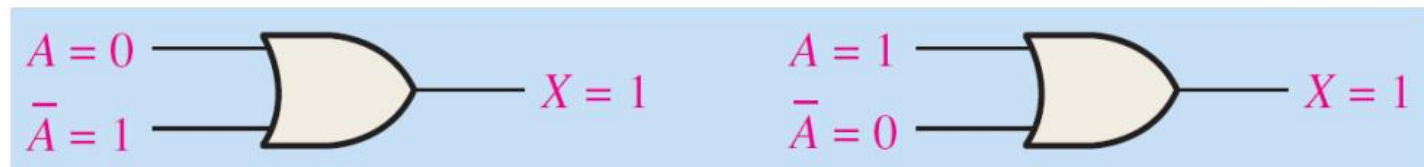


Rules of Boolean Algebra

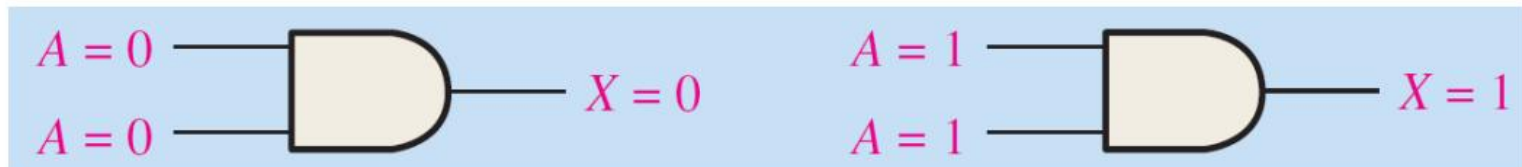
- Rule 5: $A + A = A$, A variable ORed with itself is always equal to the variable.



- Rule 6: $A + \bar{A} = 1$, A variable ORed with its complement is always equal to 1.



- Rule 7: $A \cdot A = A$, A variable ANDed with itself is always equal to the variable.

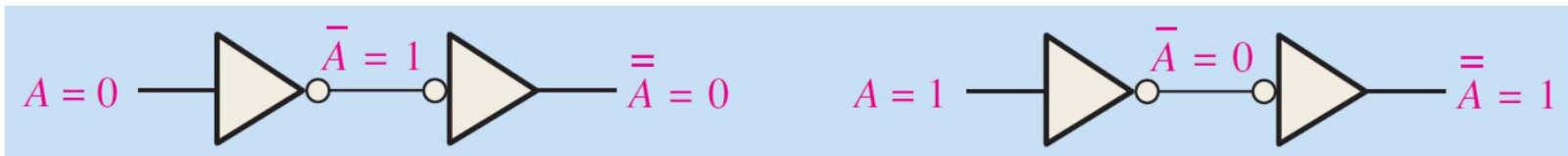


- Rule 8: $A \cdot \bar{A} = 0$, A variable ANDed with its complement is always equal to 0.



Rules of Boolean Algebra

- Rule 9: The double complement of a variable is always equal to the variable.



- Rule 10: $A + AB = A$



Rules of Boolean Algebra

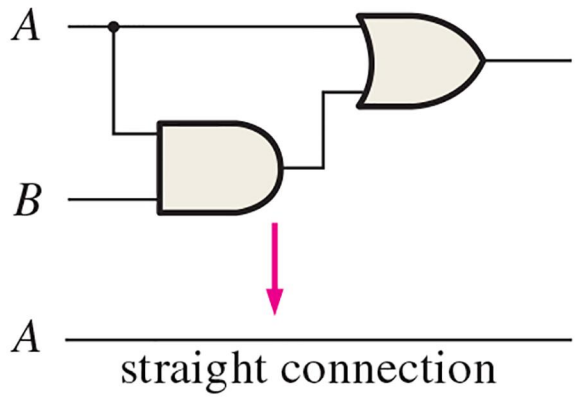
Rule 10: $A + AB = A$ This rule can be proved by applying the distributive law, rule 2, and rule 4 as follows:

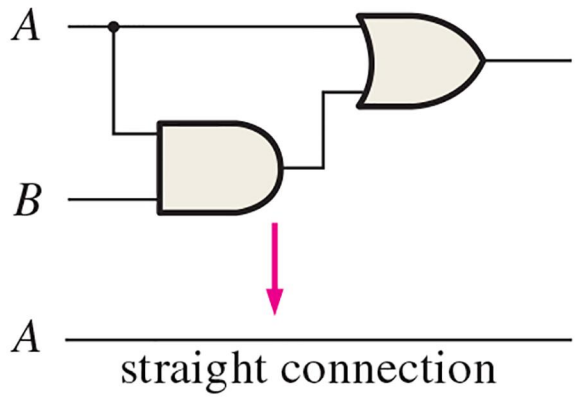
$$\begin{aligned}
 A + AB &= A \cdot 1 + AB = A(1 + B) && \text{Factoring (distributive law)} \\
 &= A \cdot 1 && \text{Rule 2: } (1 + B) = 1 \\
 &= A && \text{Rule 4: } A \cdot 1 = A
 \end{aligned}$$

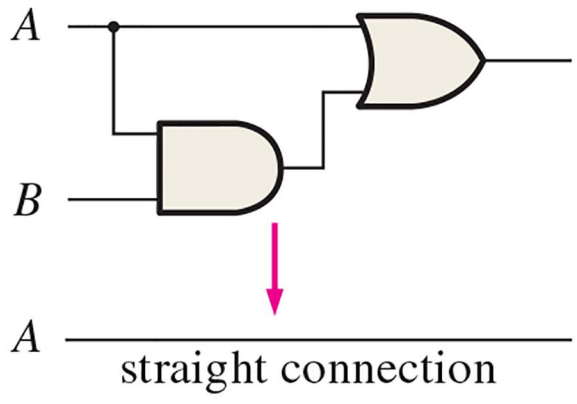
TABLE 4-2

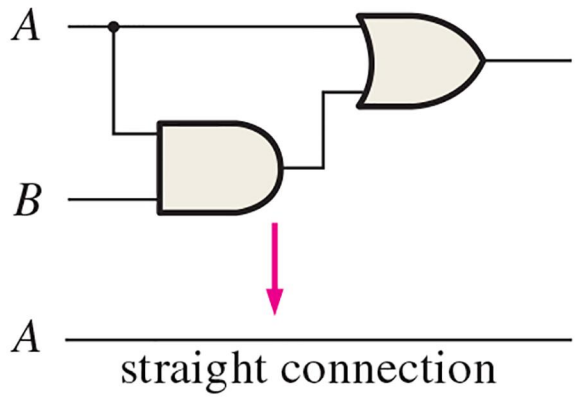
Rule 10: $A + AB = A$. Open file T04-02 to verify.

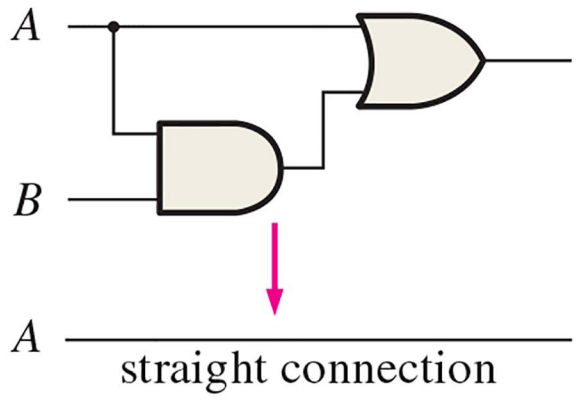
A	B	AB	$A + AB$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

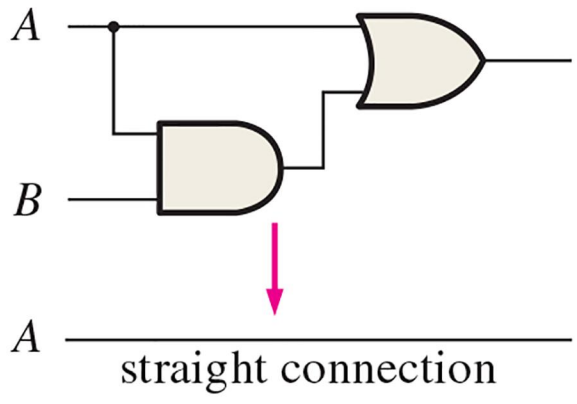


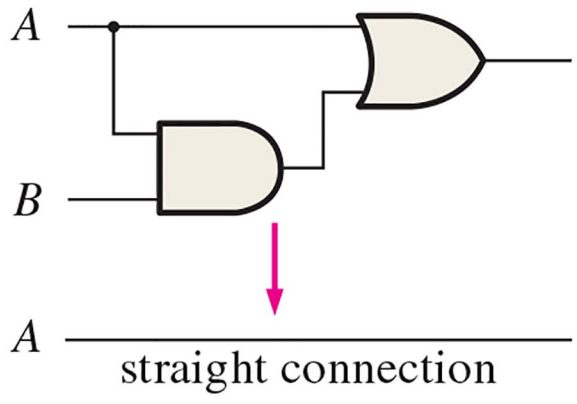


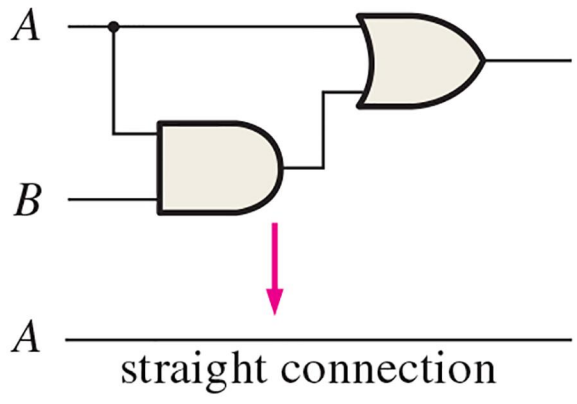


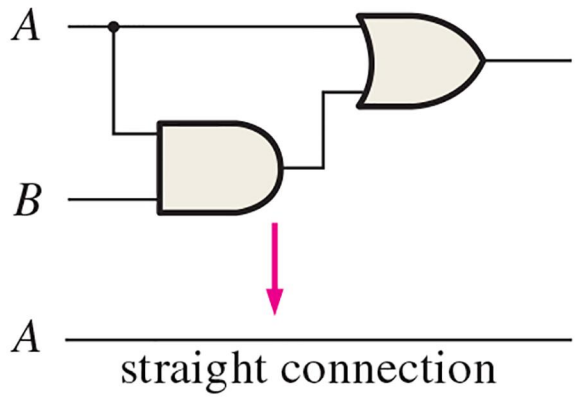


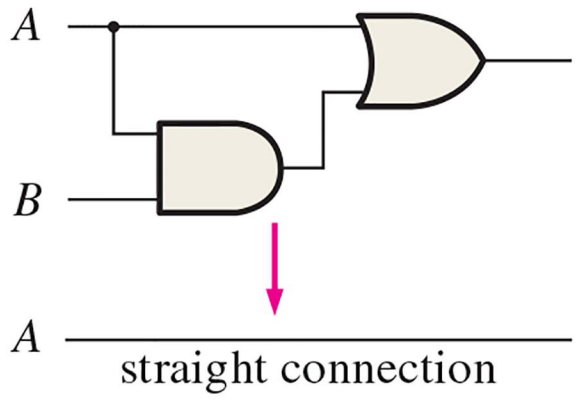


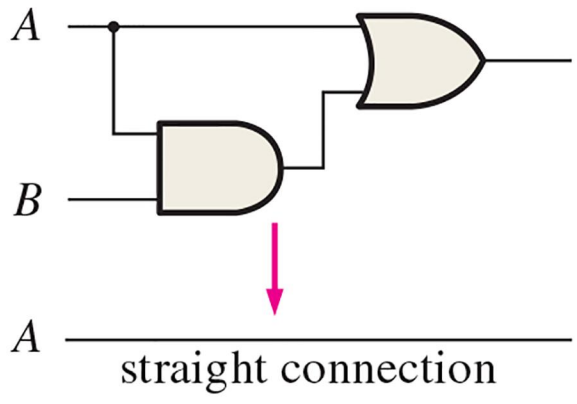


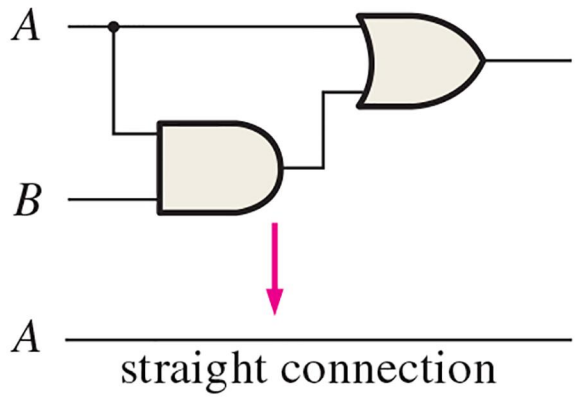


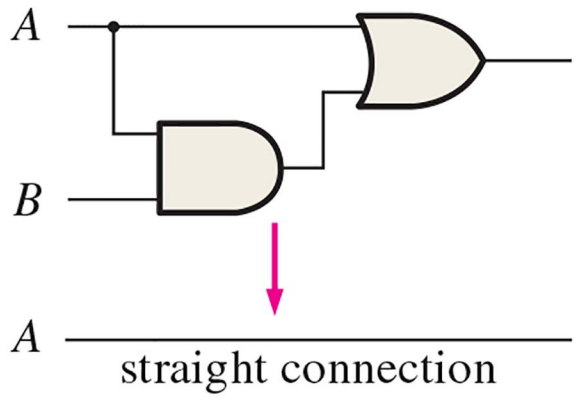


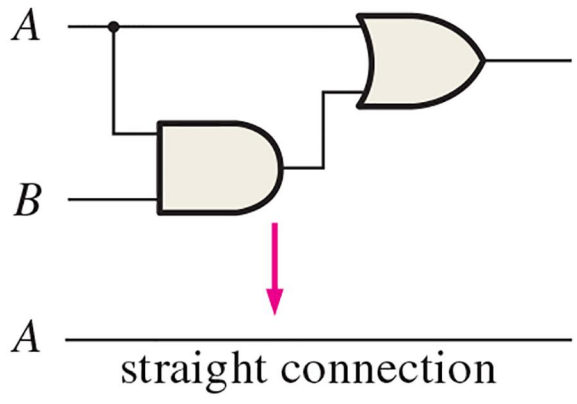


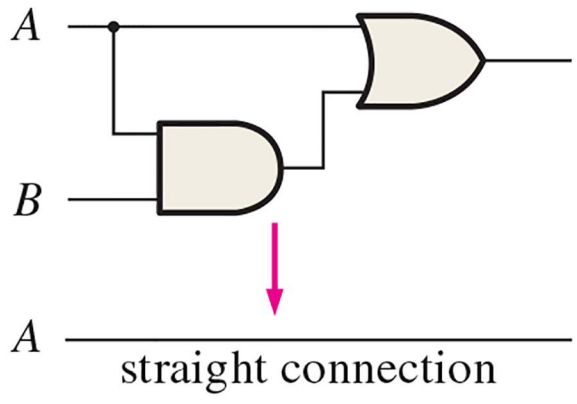


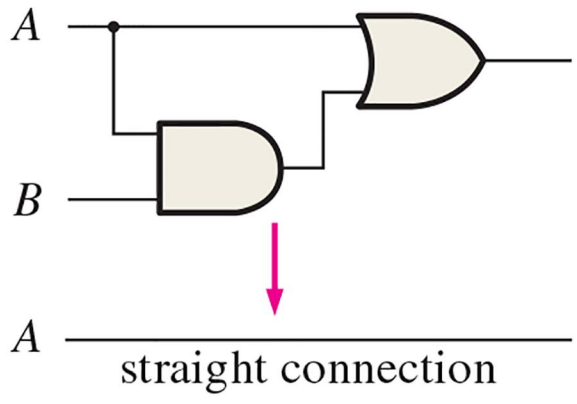


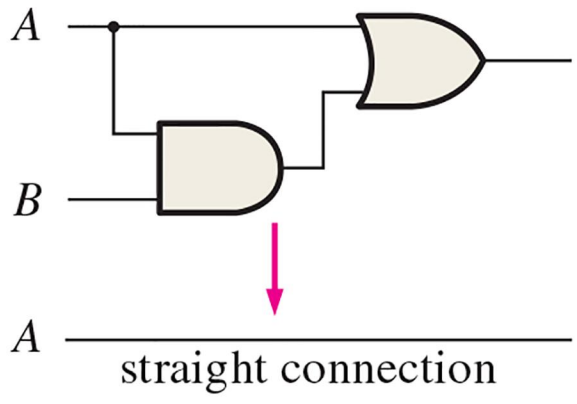


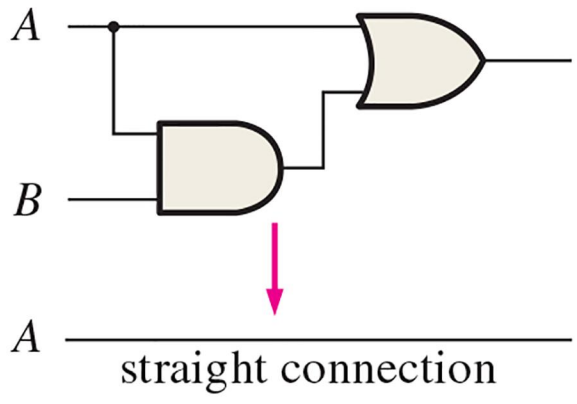


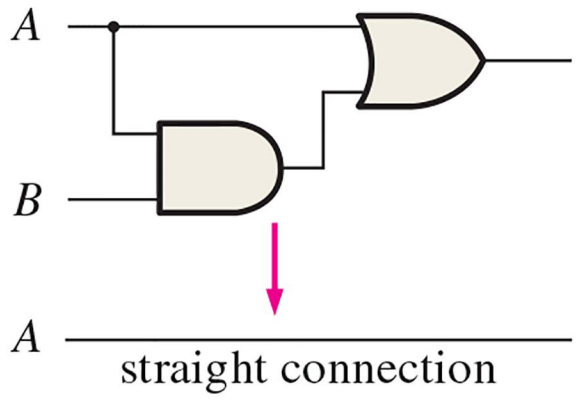


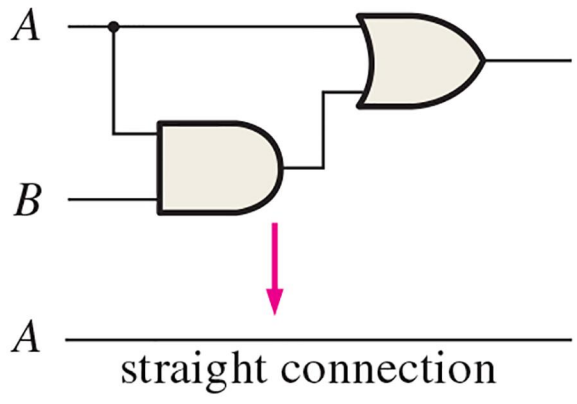


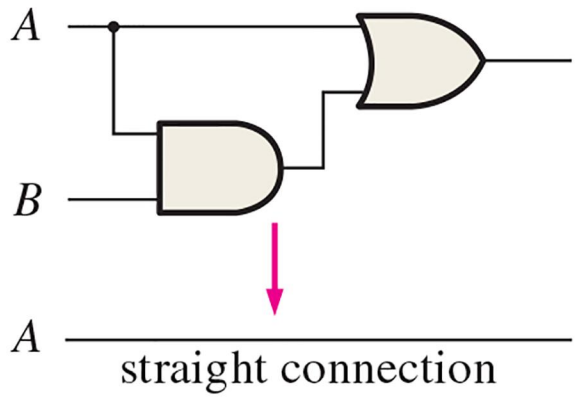


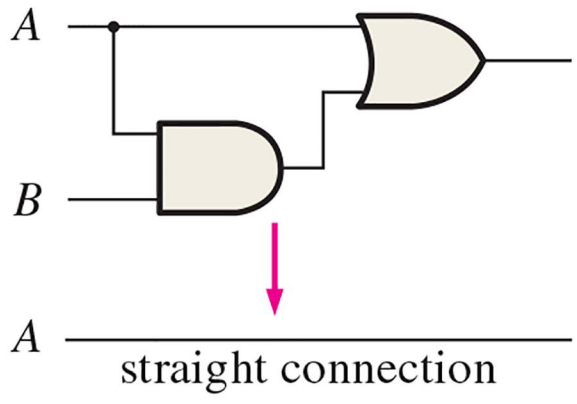


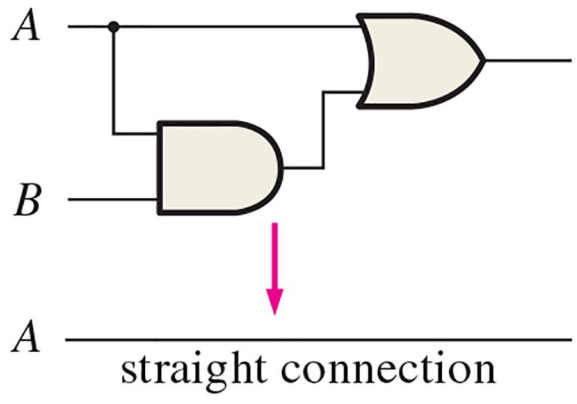


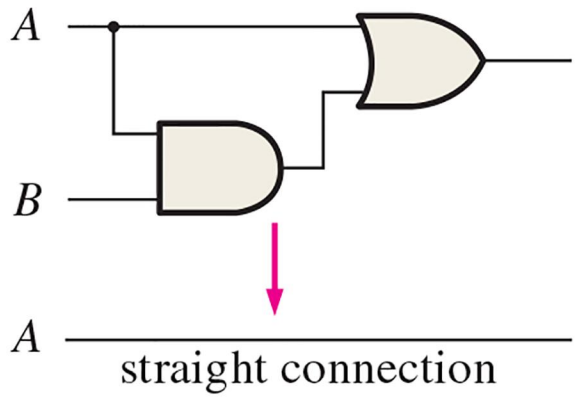


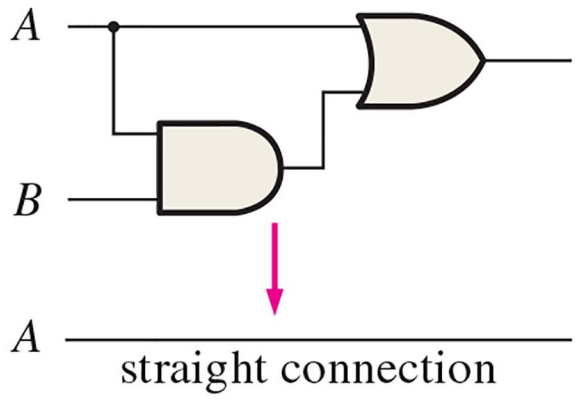


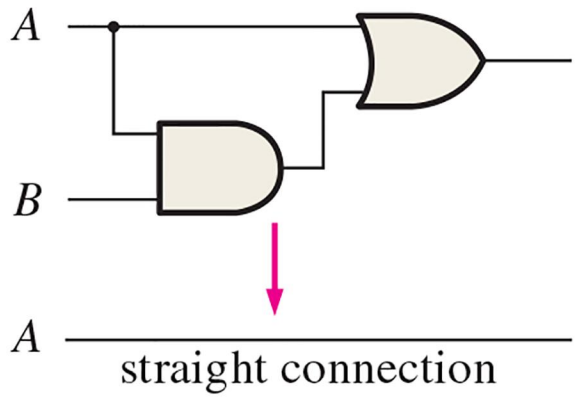


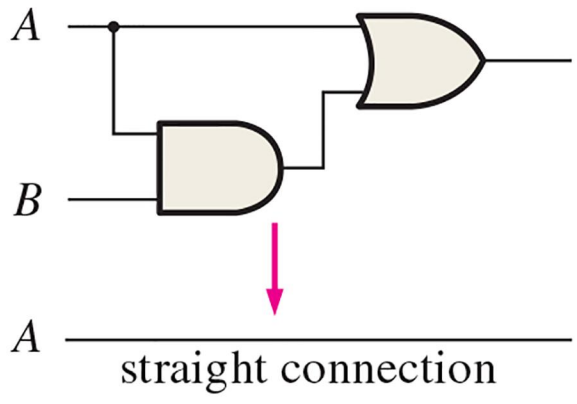


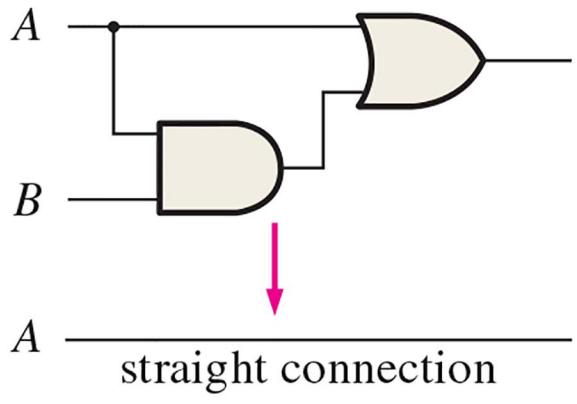


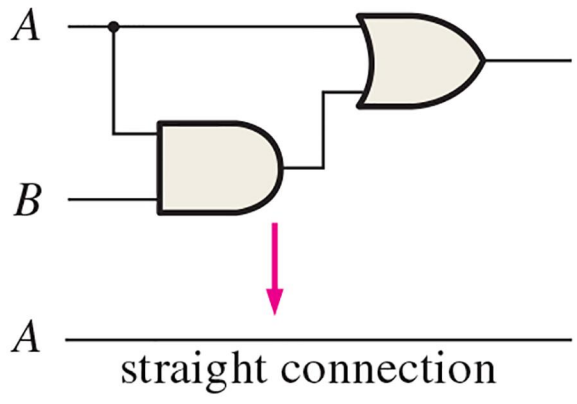


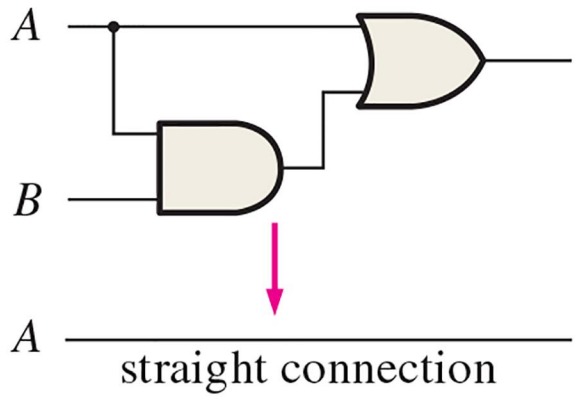


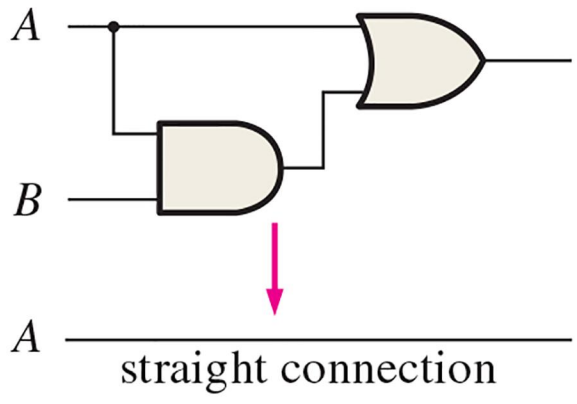


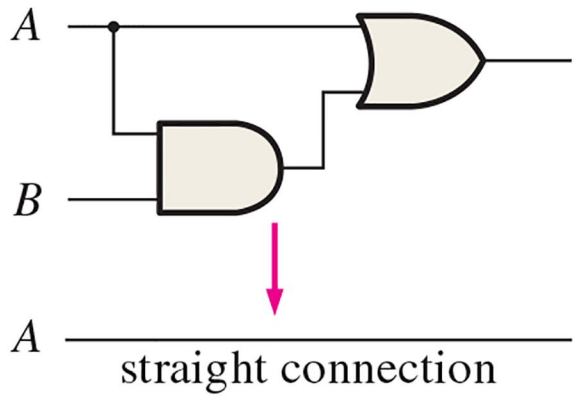


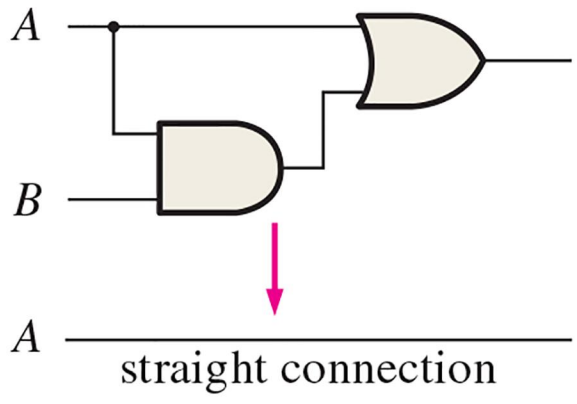


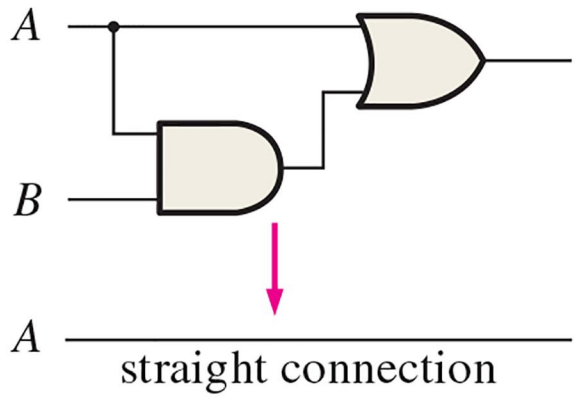


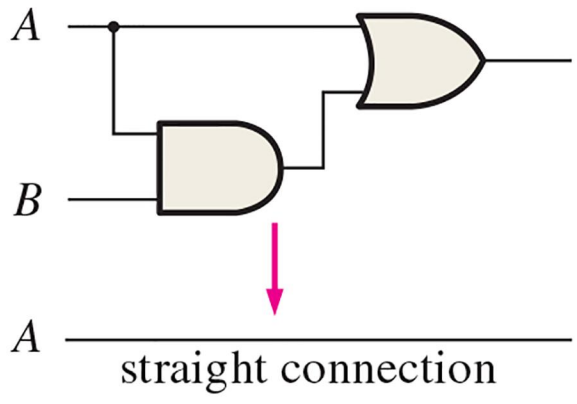


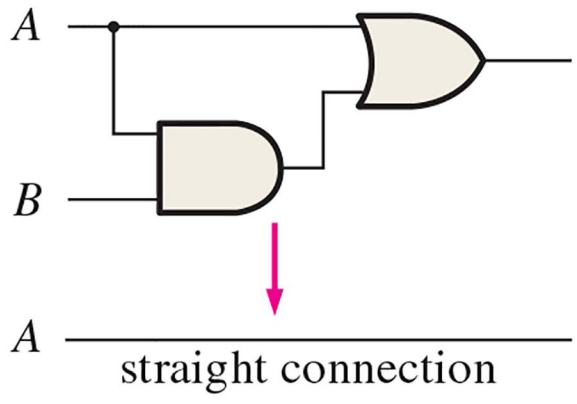


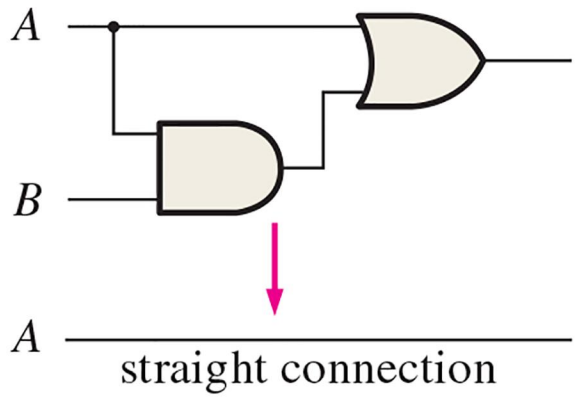


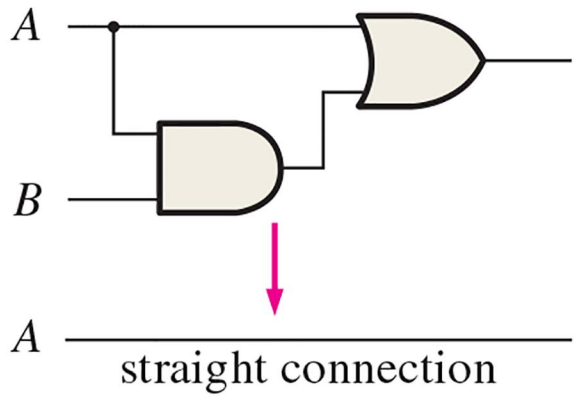


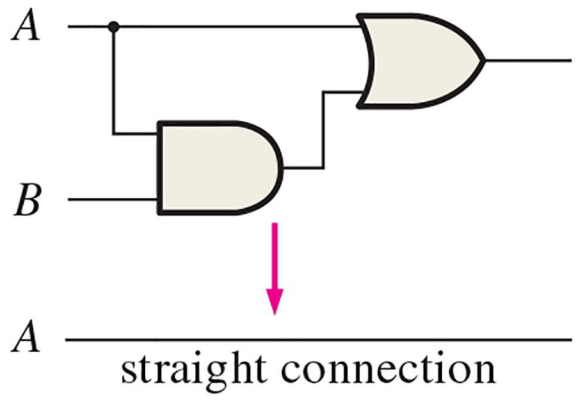


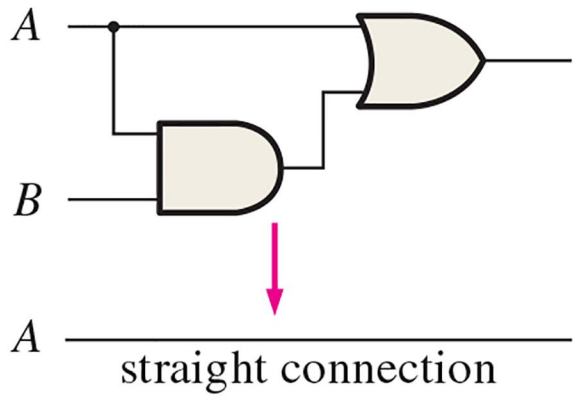


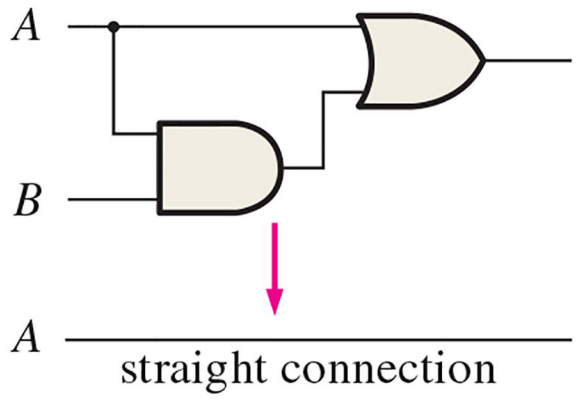


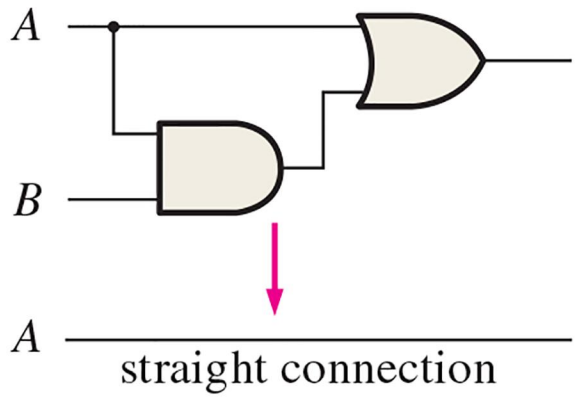


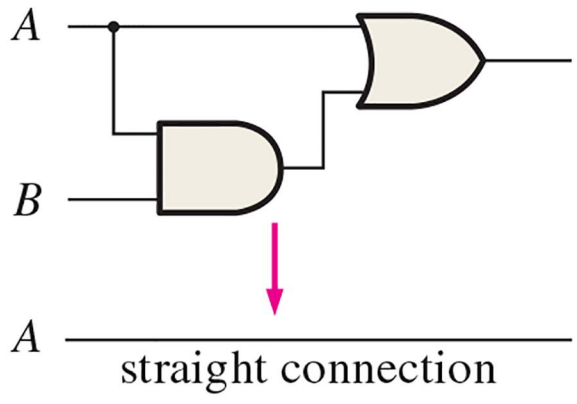


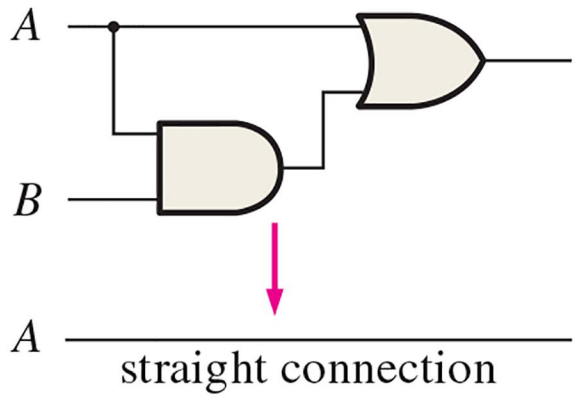


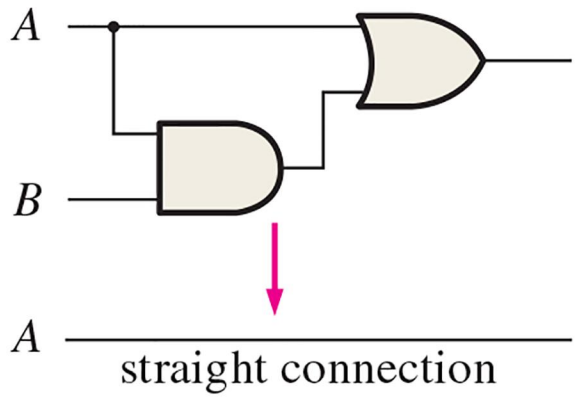


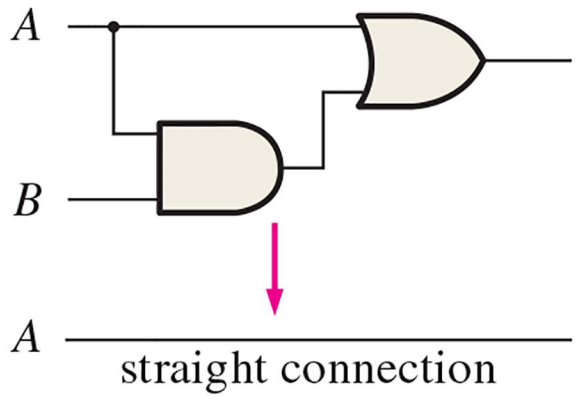


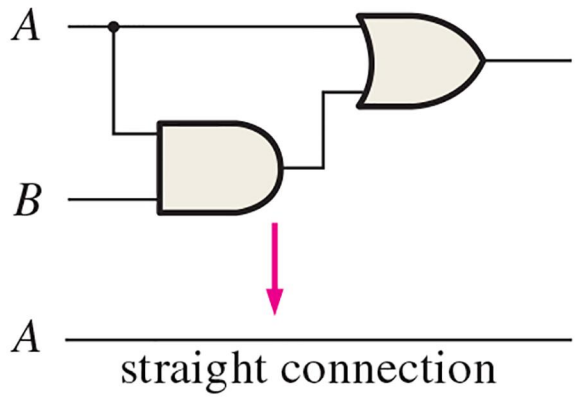


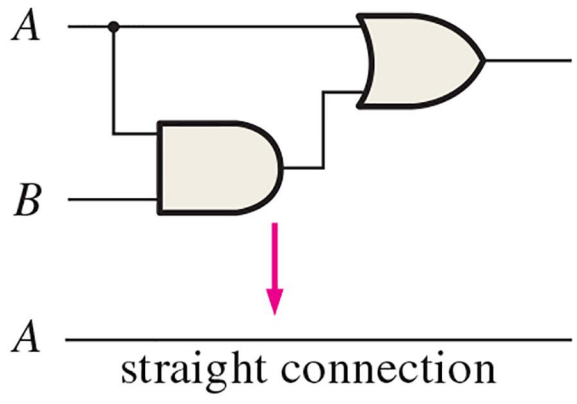


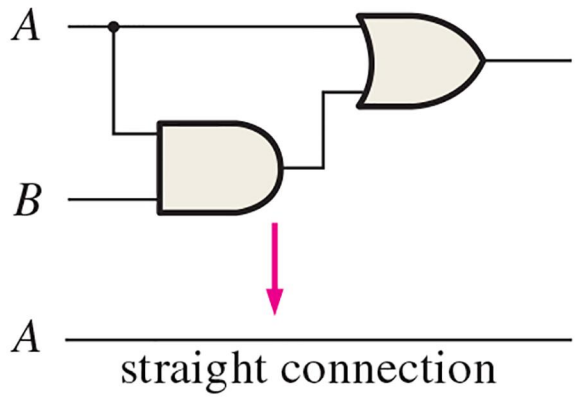


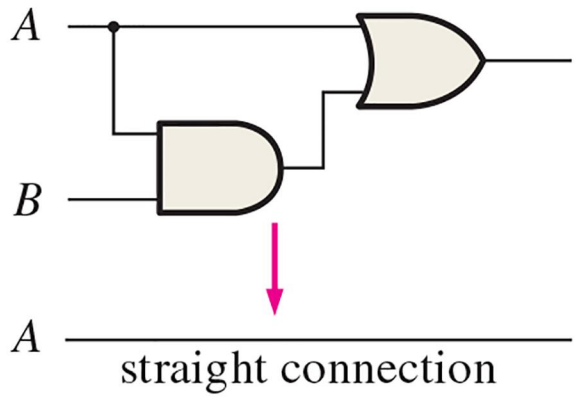


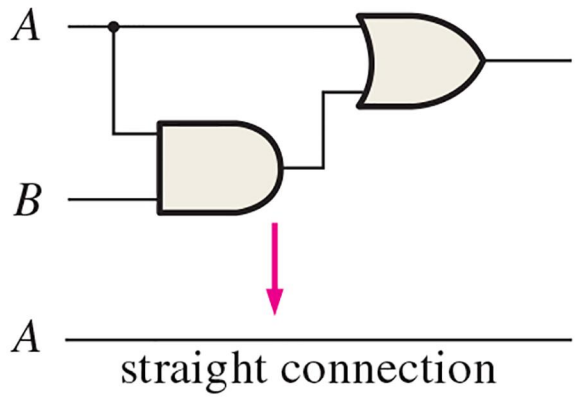


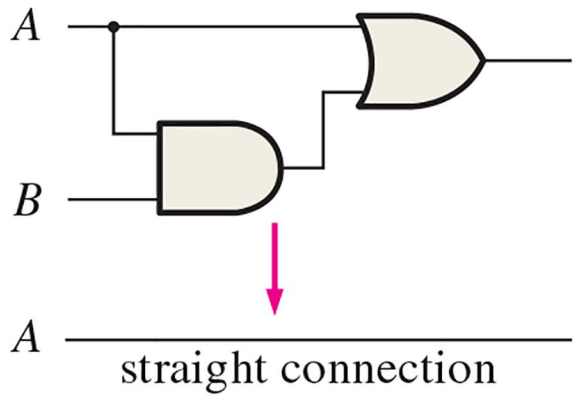


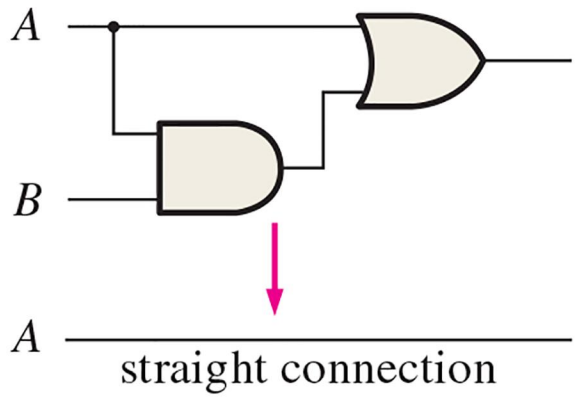


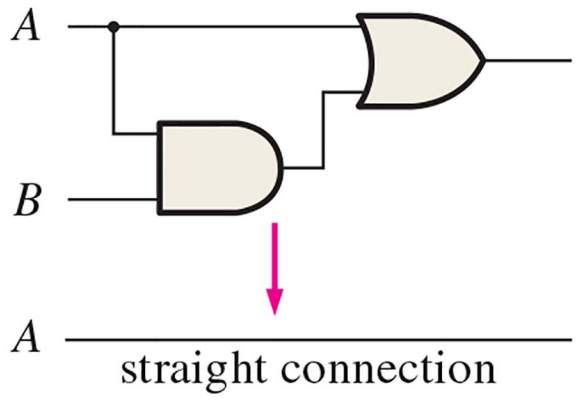


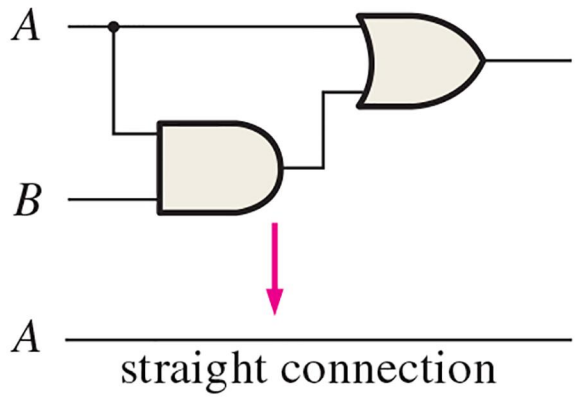


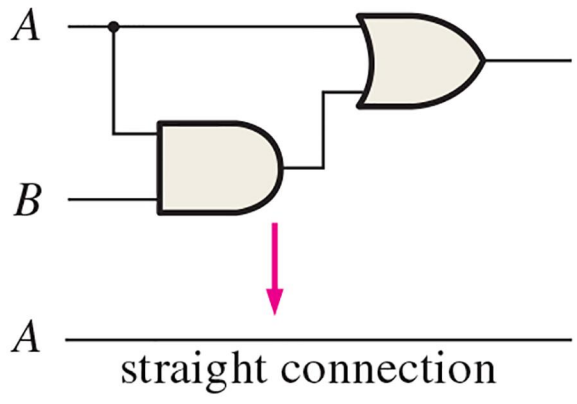


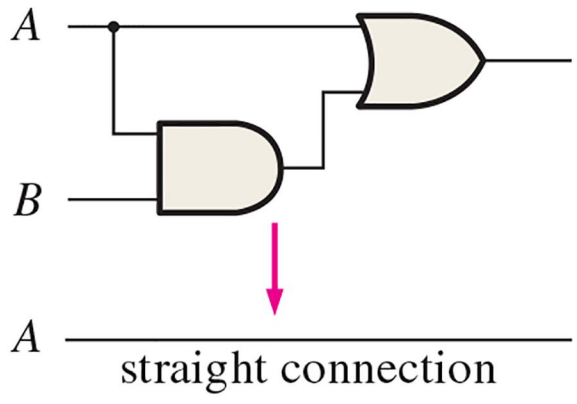


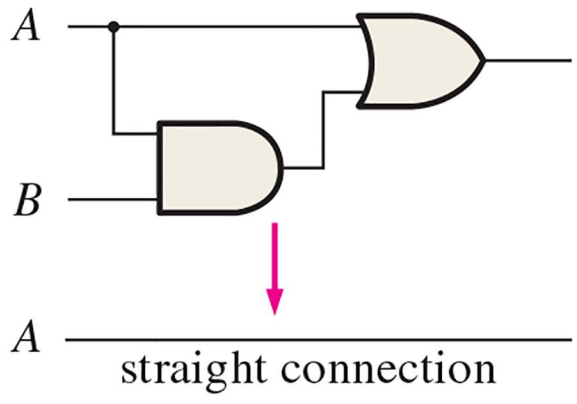


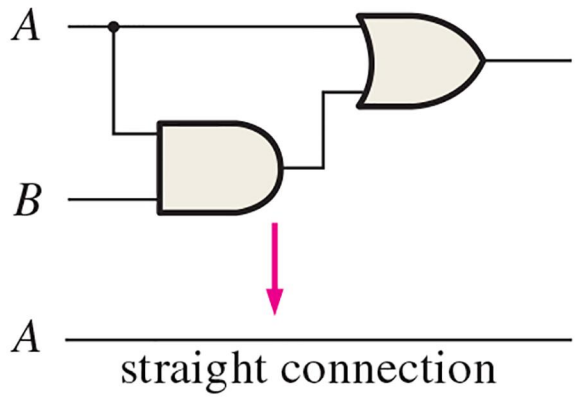


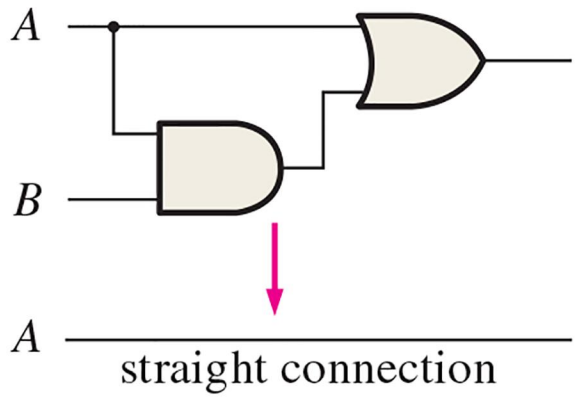


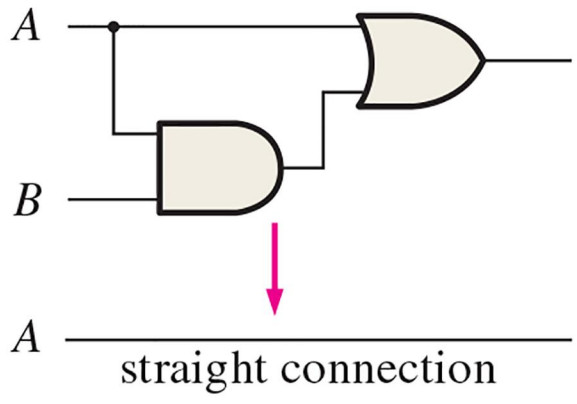


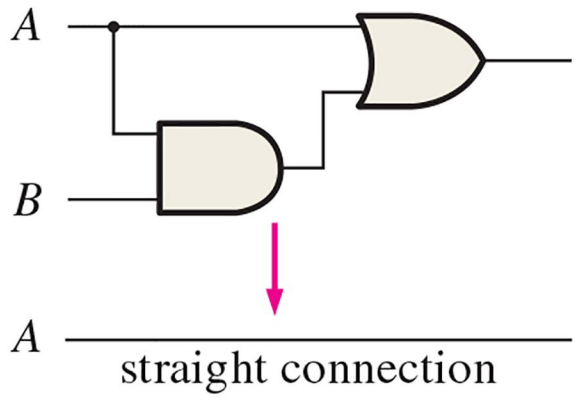


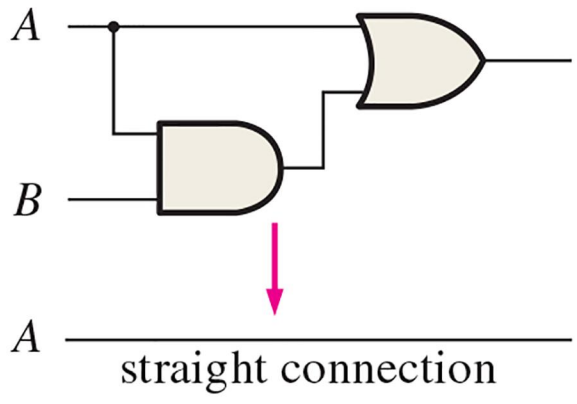


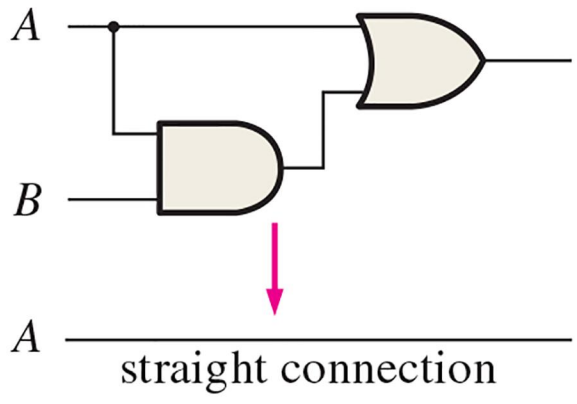


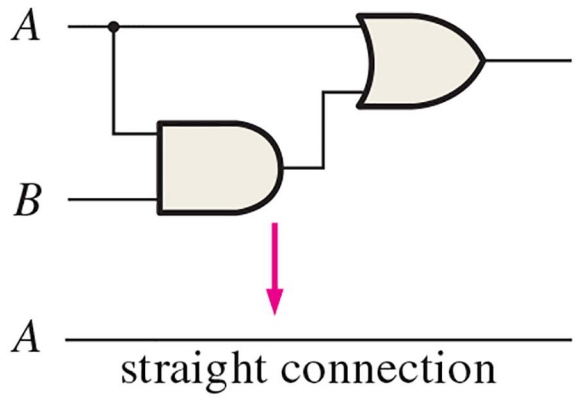


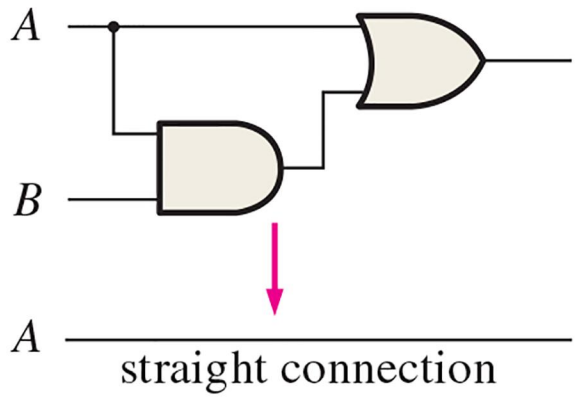


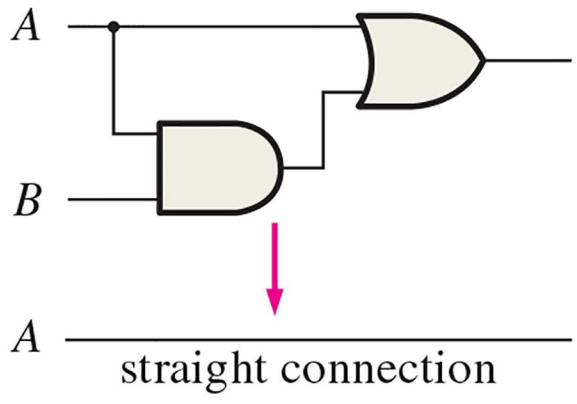


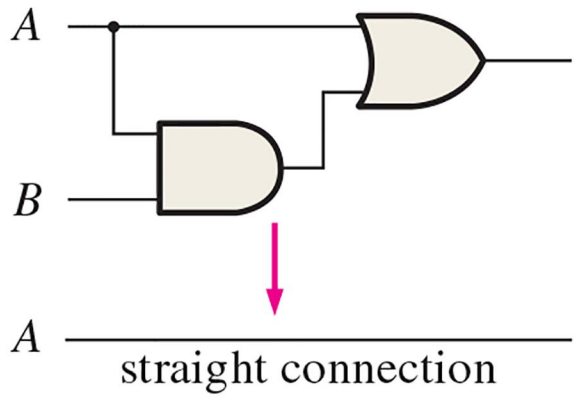


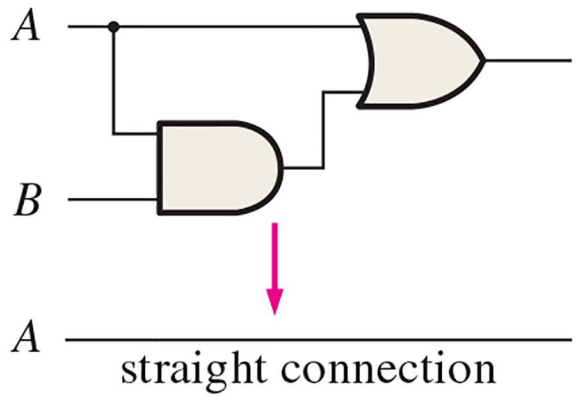


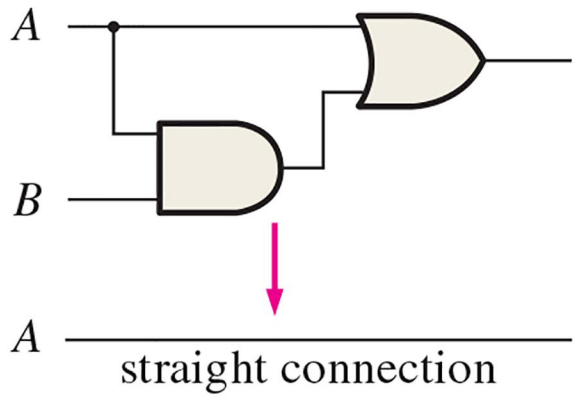


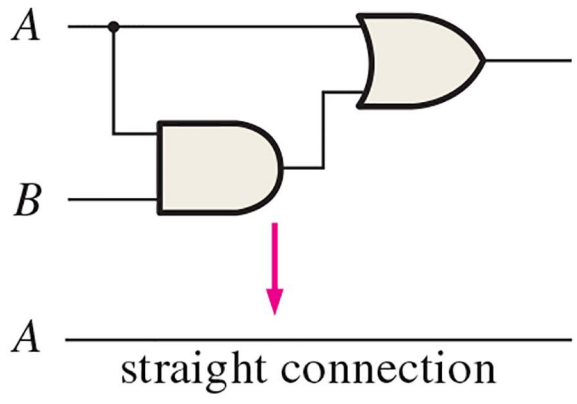


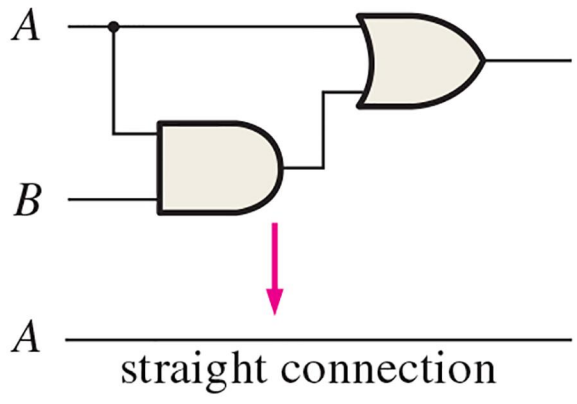


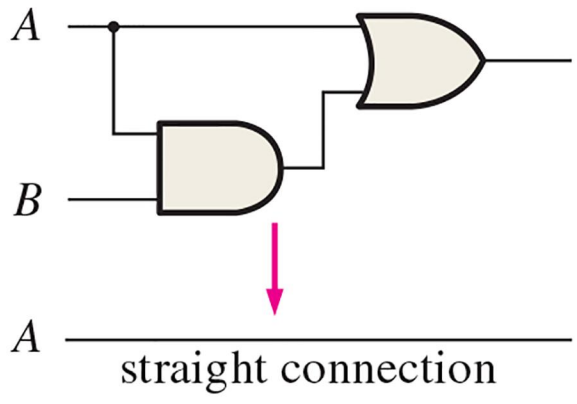


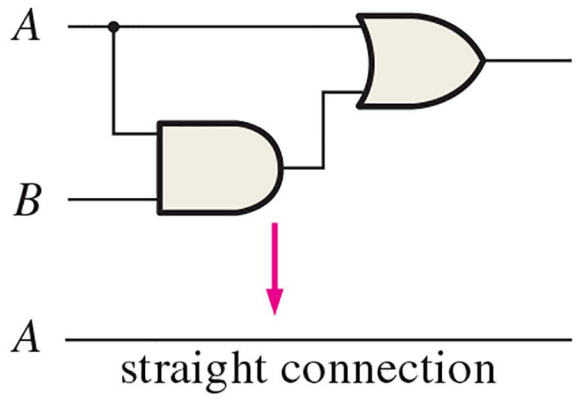


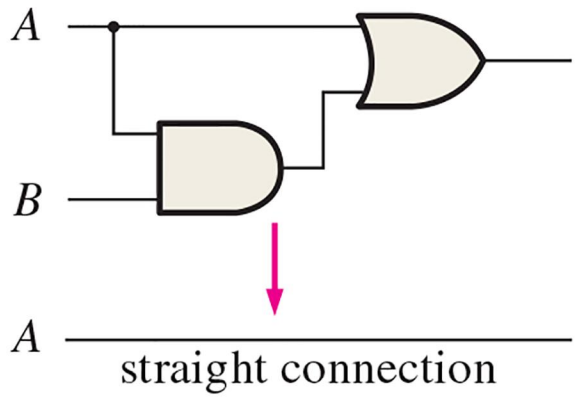


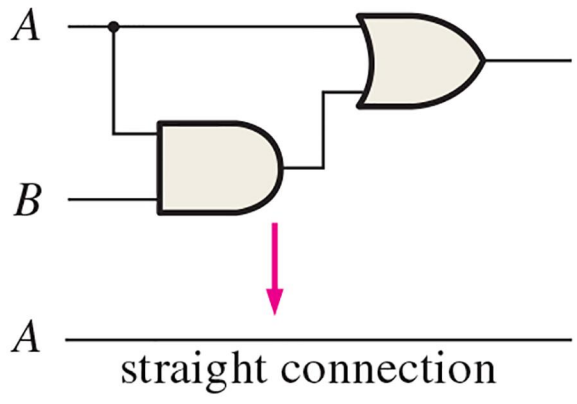


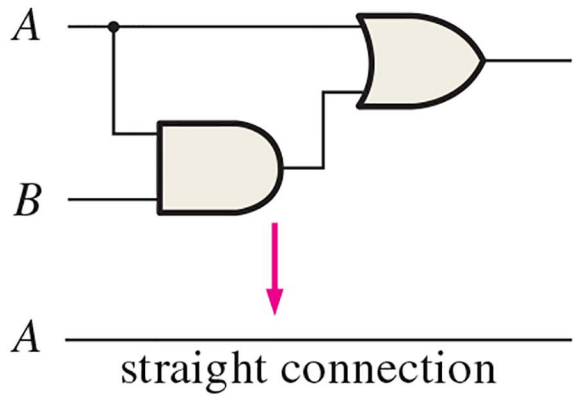


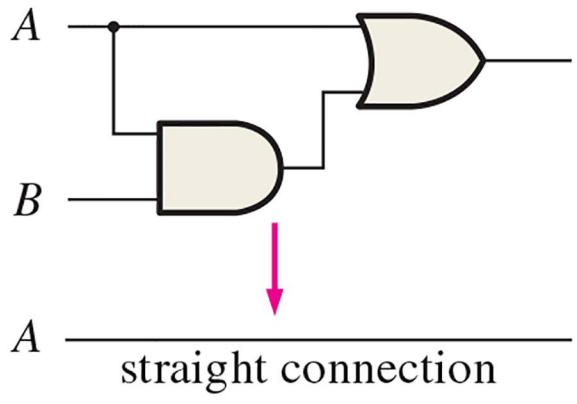


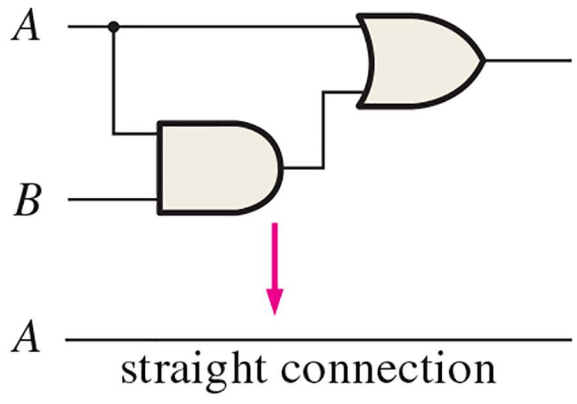


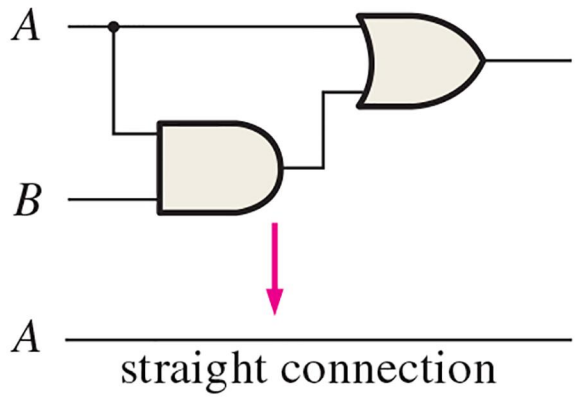


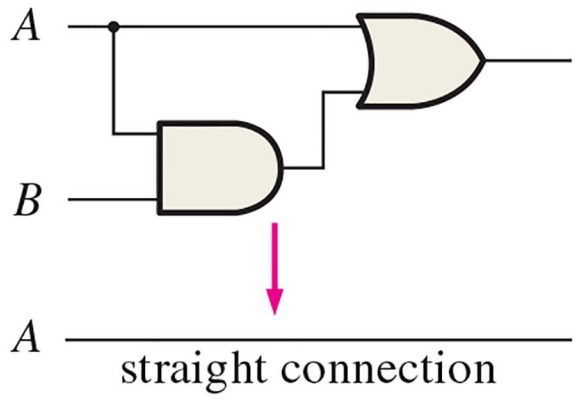


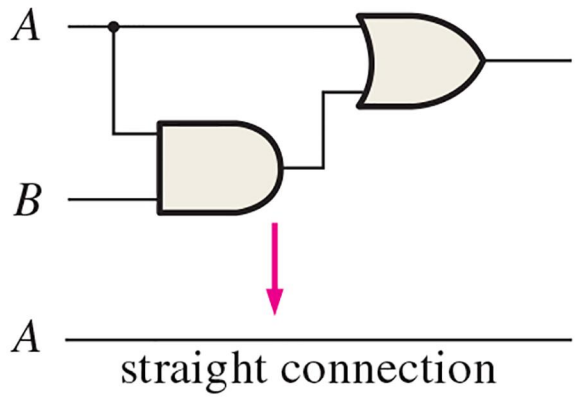


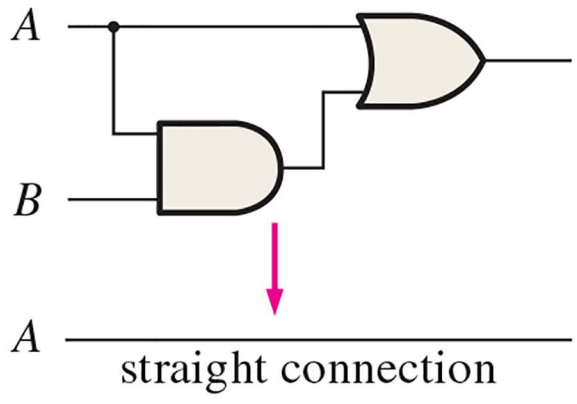


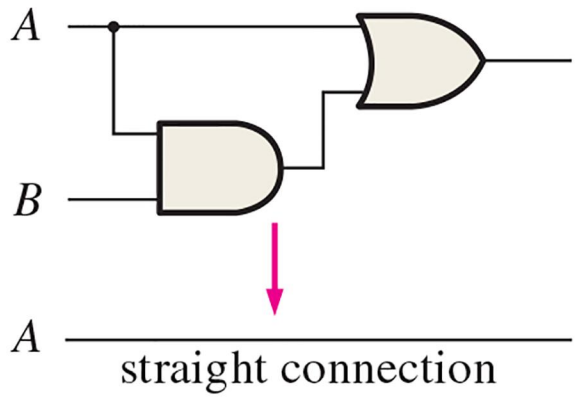


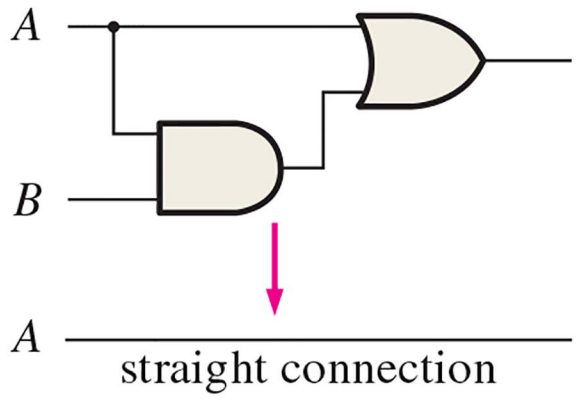


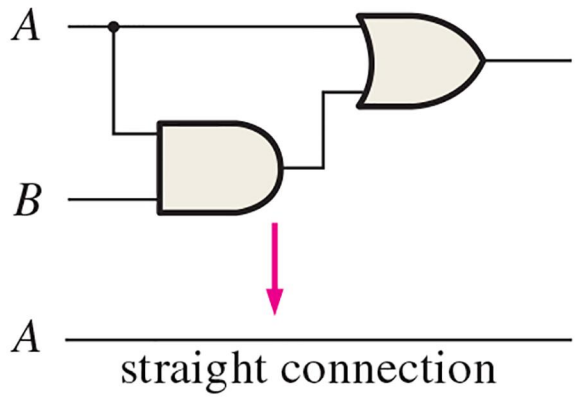


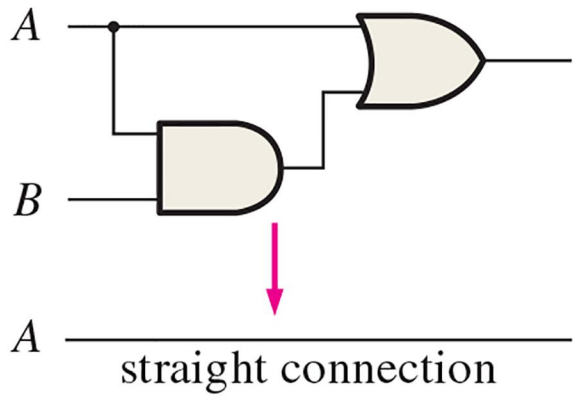


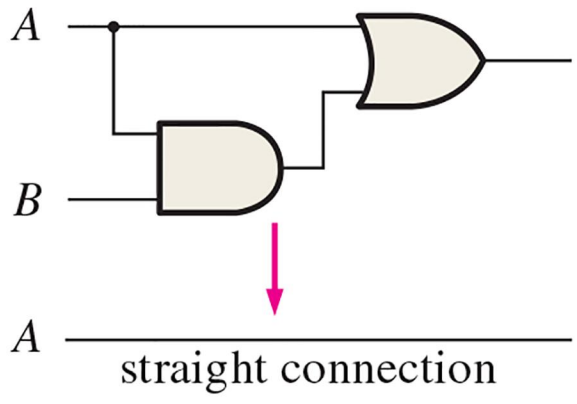


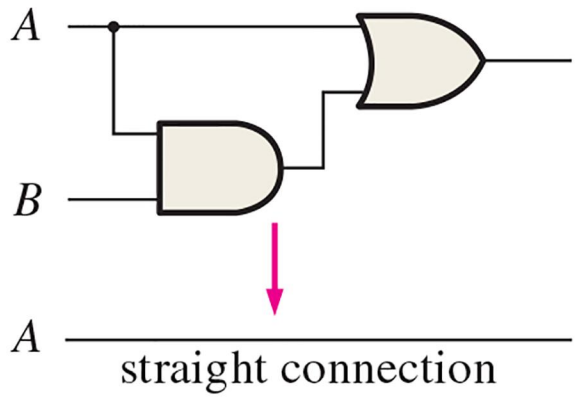


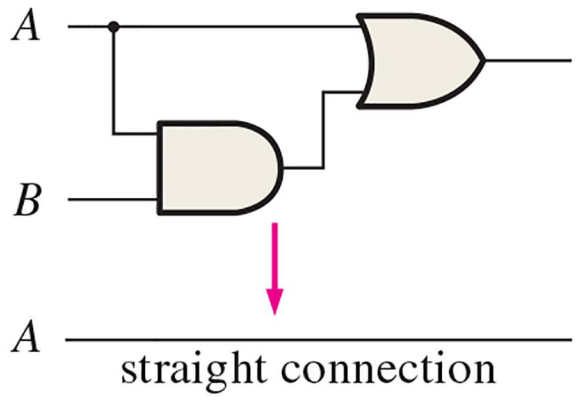


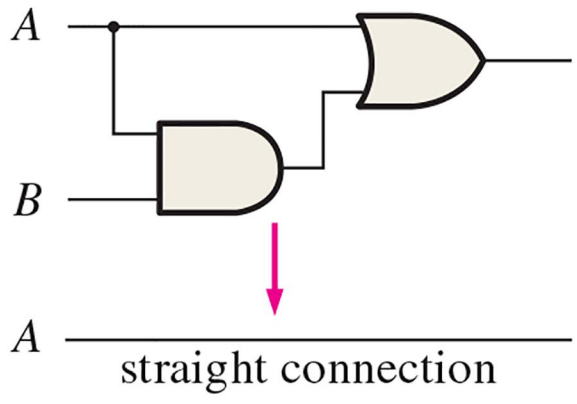


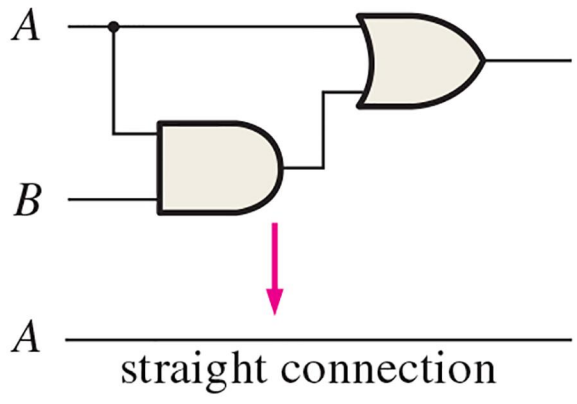


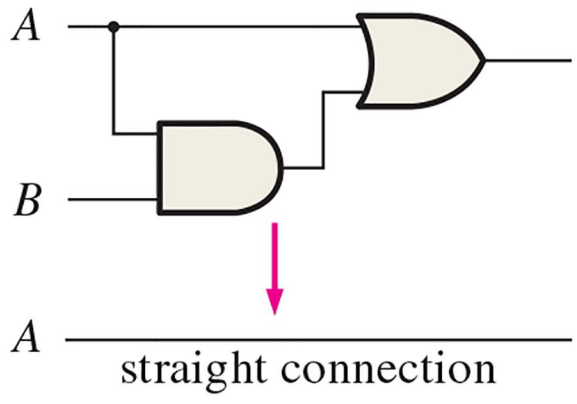


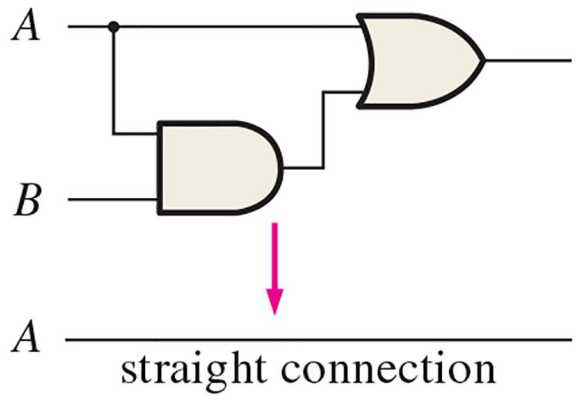


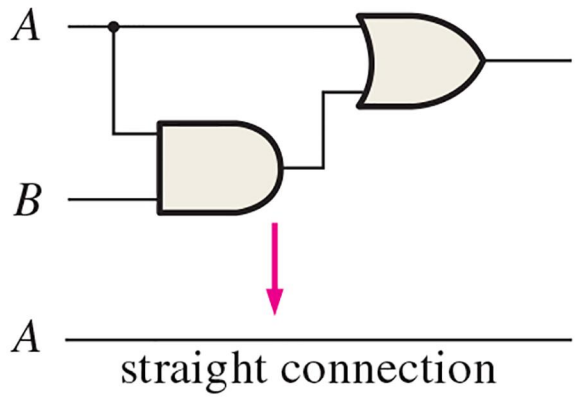


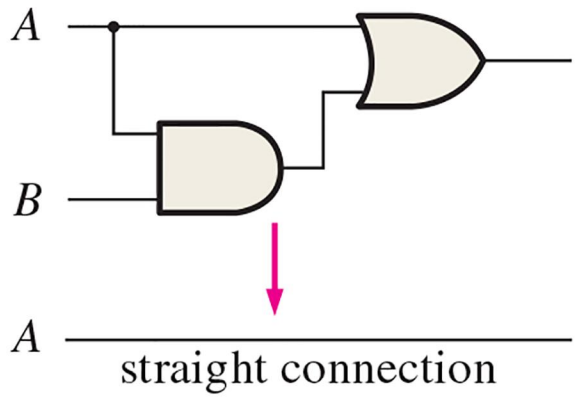


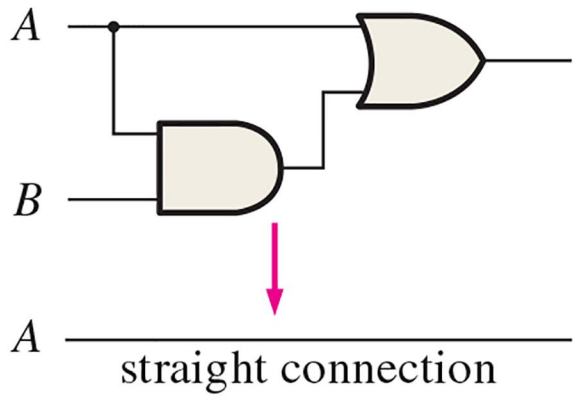


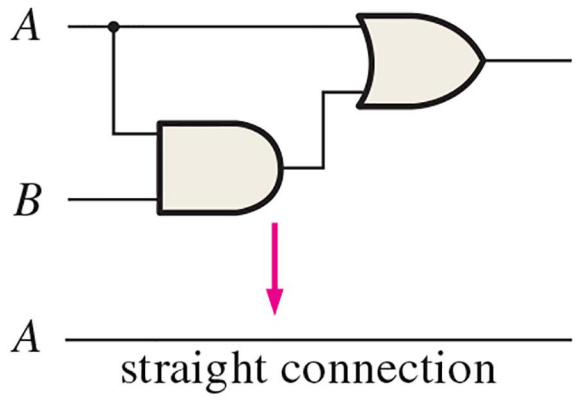


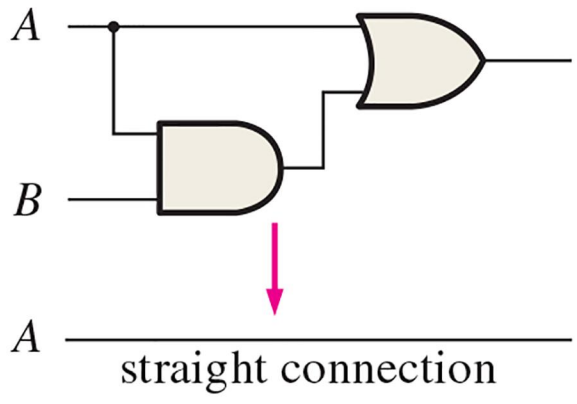


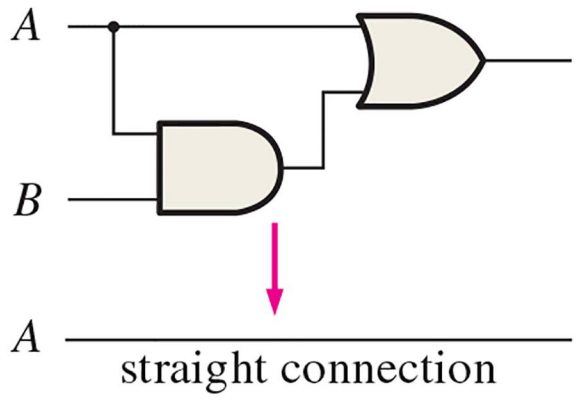


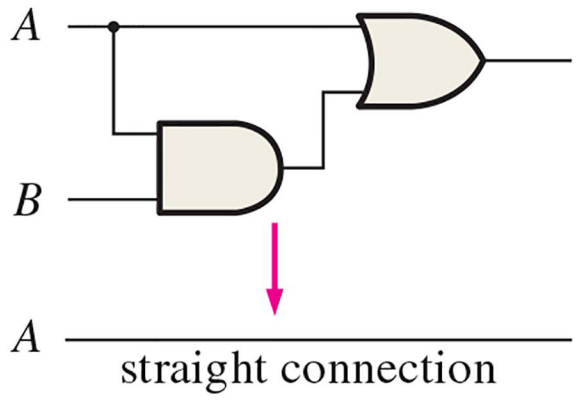


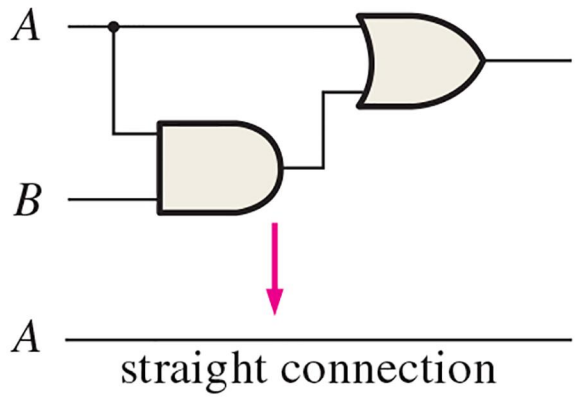


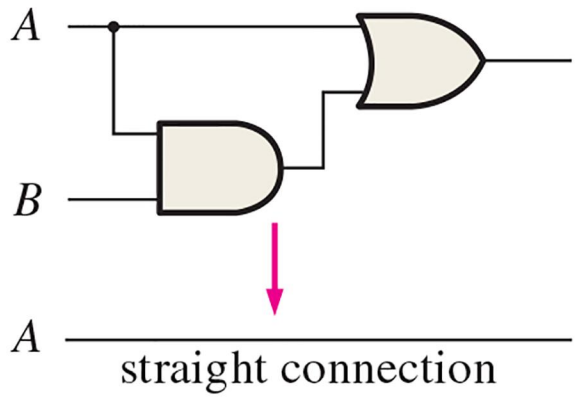


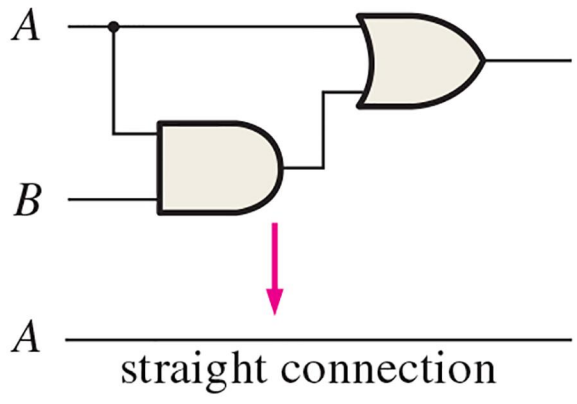


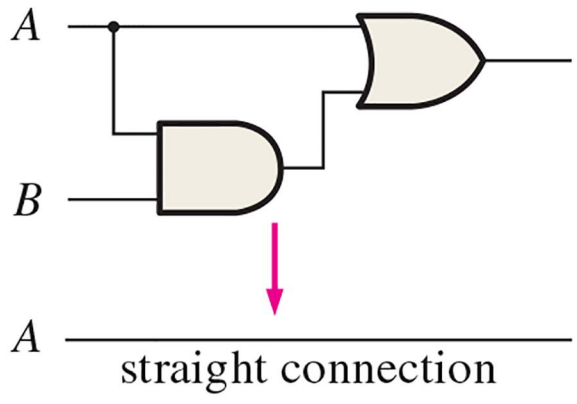


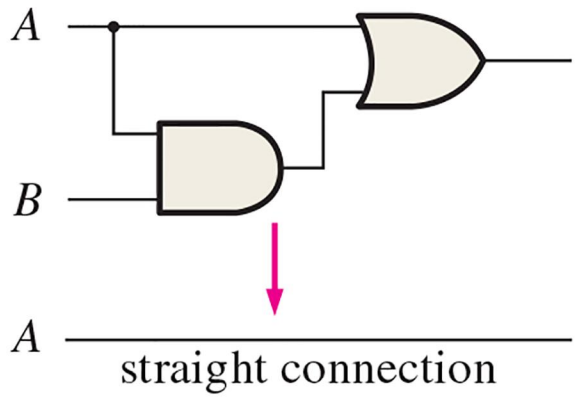


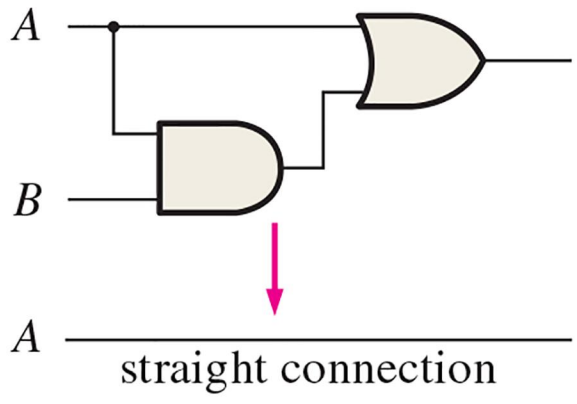


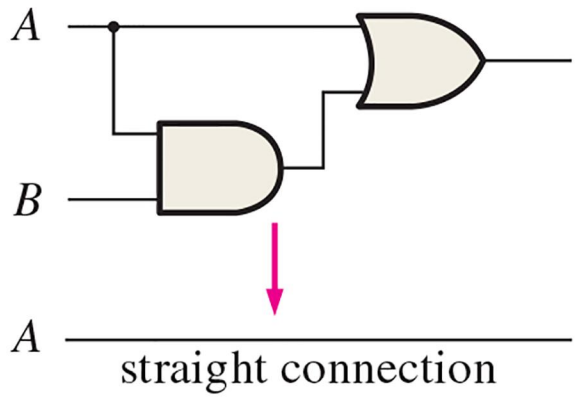


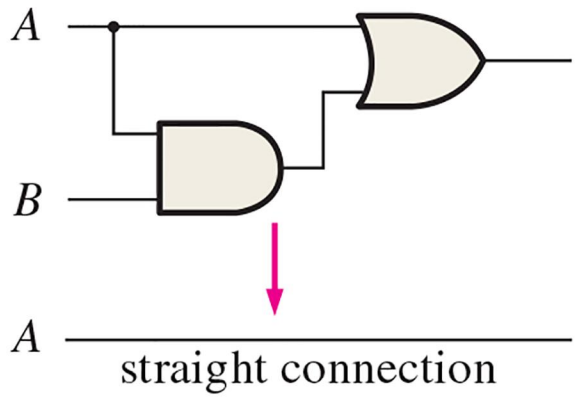


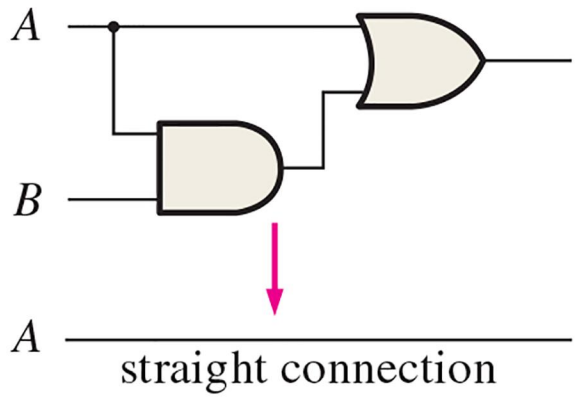


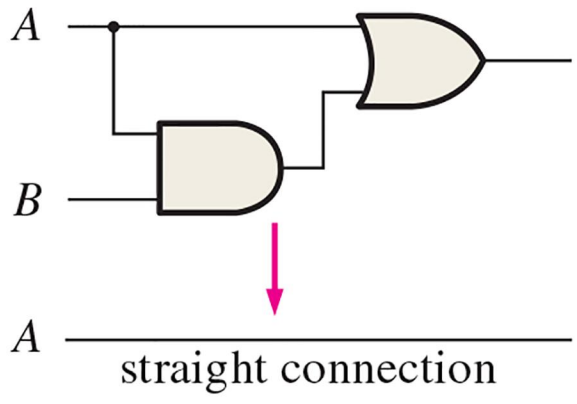


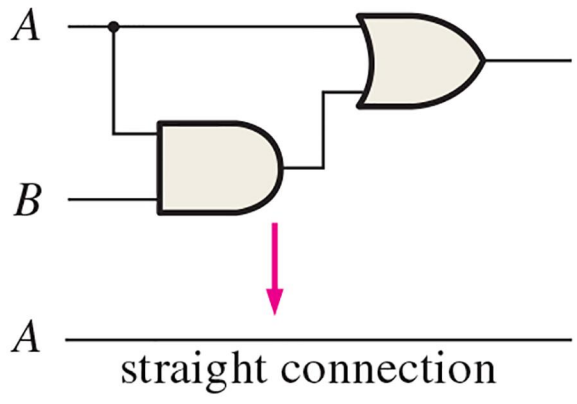


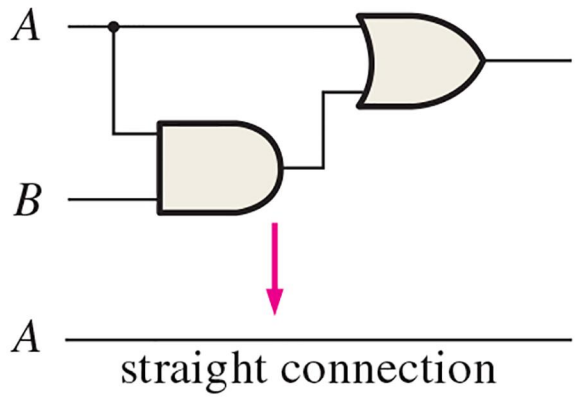


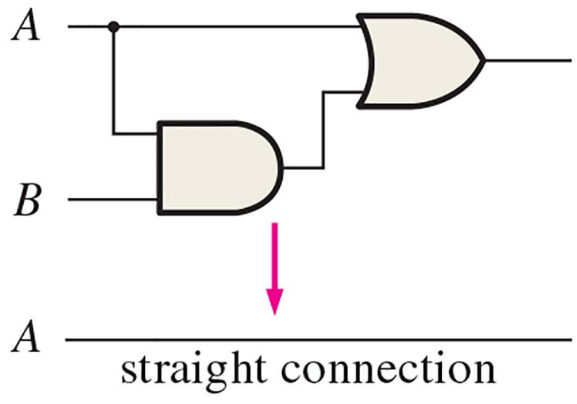


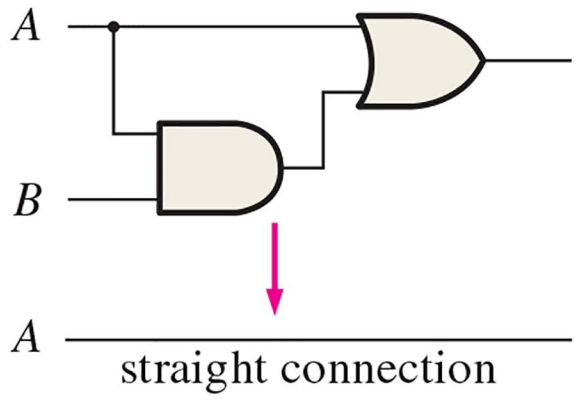


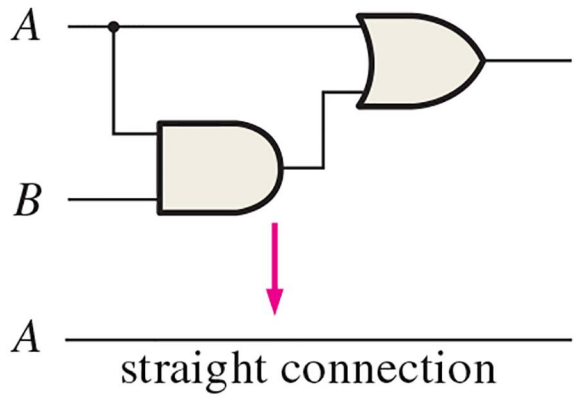


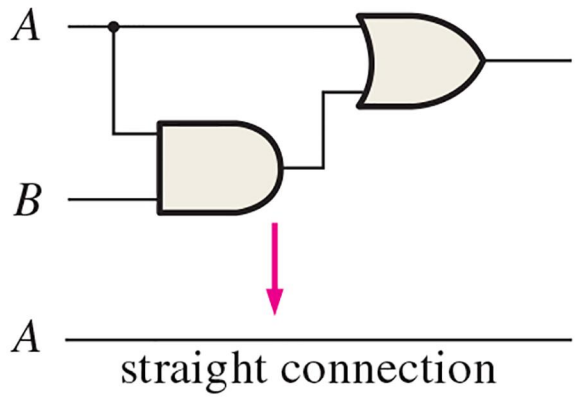


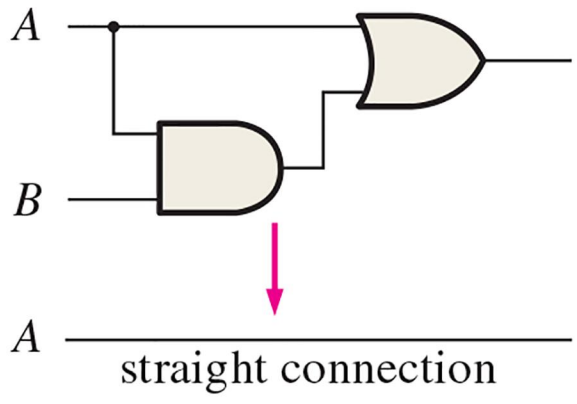


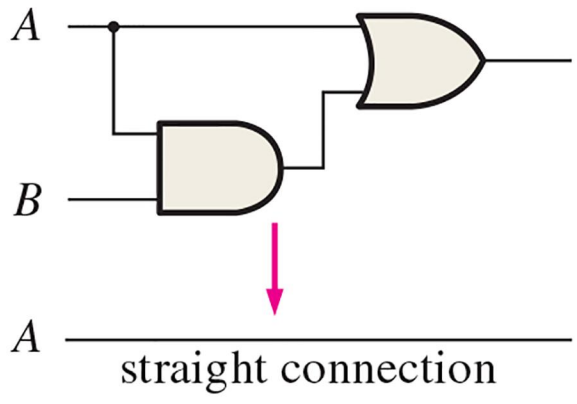


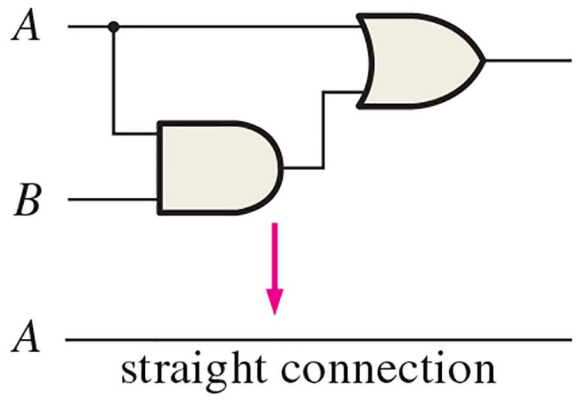


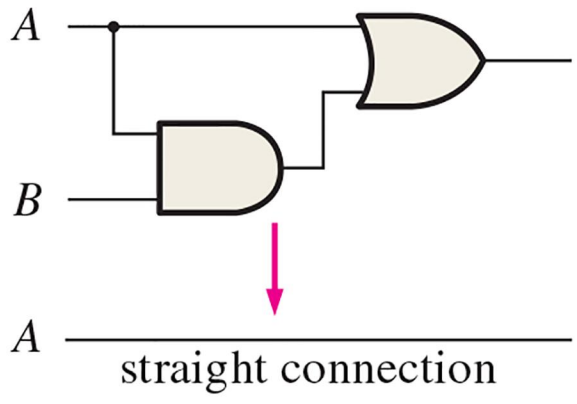


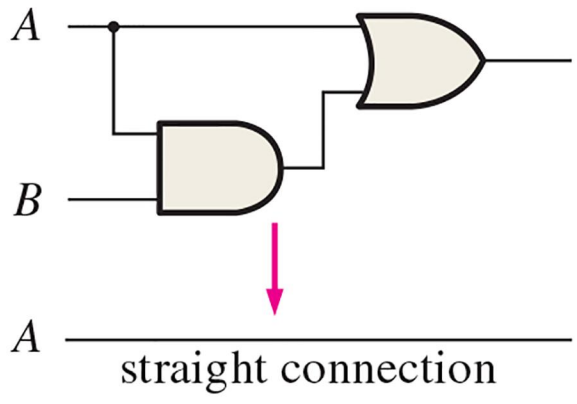


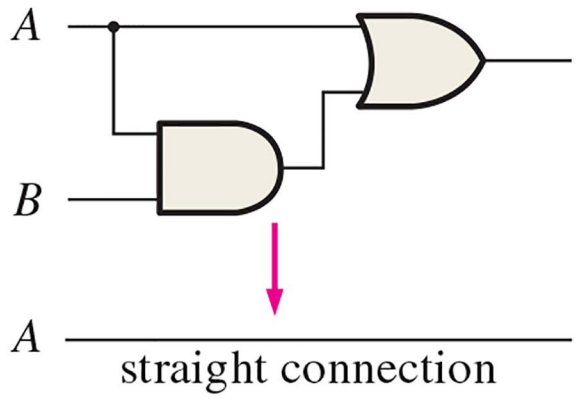


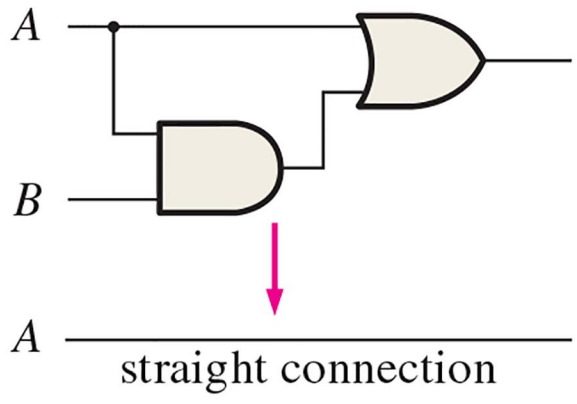


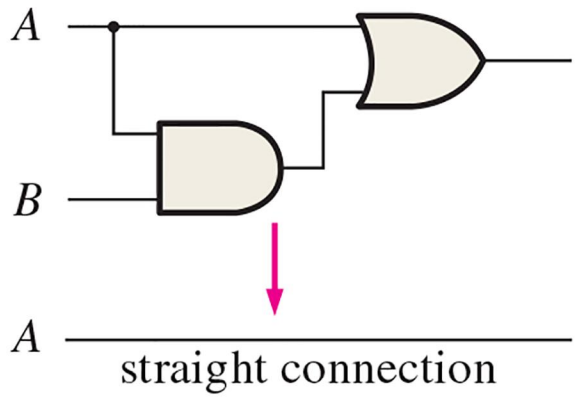












Rules of Boolean Algebra

Exercises: Prove

- Rule 11: $A + A'B = A + B$
- Rule 12: $(A + B)(A + C) = A + BC$



Rules of Boolean Algebra

- Rule 11: $A + A'B = A + B$

$$\begin{aligned}A + \bar{A}B &= (A + AB) + \bar{A}B \\&= (AA + AB) + \bar{A}B \\&= AA + AB + A\bar{A} + \bar{A}B \\&= (A + \bar{A})(A + B) \\&= 1 \cdot (A + B) \\&= A + B\end{aligned}$$

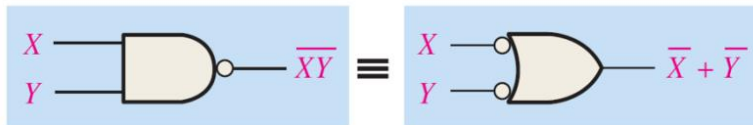
- Rule 12: $(A + B)(A + C) = A + BC$

$$\begin{aligned}(A + B)(A + C) &= AA + AC + AB + BC \\&= A + AC + AB + BC \\&= A(1 + C) + AB + BC \\&= A \cdot 1 + AB + BC \\&= A(1 + B) + BC \\&= A \cdot 1 + BC \\&= A + BC\end{aligned}$$



DeMorgan's Theorems

- DeMorgan's first theorem $\overline{XY} = \overline{X} + \overline{Y}$



NAND

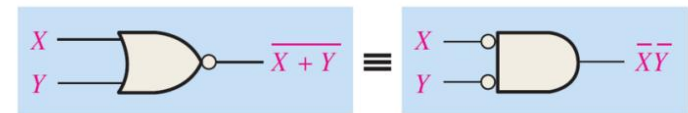
Negative-OR

Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

Think about it:

- 3 variable DeMorgan's Theorems?

- DeMorgan's second theorem $\overline{\overline{X} + \overline{Y}} = \overline{\overline{X}} \overline{\overline{Y}}$



NOR

Negative-AND

Inputs		Output	
X	Y	$\overline{\overline{X} + \overline{Y}}$	$\overline{\overline{X}} \overline{\overline{Y}}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

- DeMorgan's theorems provide mathematical equivalency of the NAND and negative-OR gates and the equivalency of the NOR and negative-AND gates



$$\bar{A}\bar{B} \stackrel{?}{=} \overline{AB}$$



Simplify boolean expression

$$\overline{AB + AC} + \overline{A}\overline{B}C$$



Simplify Boolean Expression

$$\begin{aligned} & \overline{AB + AC} + \overline{A} \overline{B} C \\ \text{DeMorgan's theorem} \quad & \downarrow \\ & (\overline{AB})(\overline{AC}) + \overline{A} \overline{B} C \\ \text{DeMorgan's theorem} \quad & \downarrow \\ & (\overline{A} + \overline{B})(\overline{A} + \overline{C}) + \overline{A} \overline{B} C \\ \text{Distributive law} \quad & \downarrow \\ & \overline{A} \overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C} + \overline{A} \overline{B} C \\ \text{Rule 7 and Rule 10} \quad & \downarrow \\ & \overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C} \end{aligned}$$

$$\begin{aligned} & \text{Rule 10} \quad \downarrow \\ & \overline{A} + \overline{A} \overline{B} + \overline{B} \overline{C} \\ & \text{Rule 10} \quad \downarrow \\ & \overline{A} + \overline{B} \overline{C} \end{aligned}$$

- Use the basic laws, rules, and theorems to simplify an expression depends on a thorough knowledge and considerable practice, not to mention a little ingenuity and cleverness.

